

MATH5003P Assignment Solutions

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Abstract

This is a collection of personal solutions to homework assigned in MATH6437P, a graduate-level introduction to symplectic topology at USTC.

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1 HOMEWORK 1

Exercise 1.1. Let V be an $\mathbb{R}$ -vector space of dimension m and ω be a non-degenerate skew-symmetric bilinear form. Prove that $m = 2n$ is even and there exists a basis $u_1, \dots, u_n, v_1, \dots, v_n$ of V such that

$$\omega(u_j, u_k) = \omega(v_j, v_k) = 0, \omega(v_j, u_k) = \delta_{jk} \text{ for } j, k = 1, \dots, n$$

where δ_{jk} is 1 if $j = k$ and 0 otherwise.

Solution. Let Ω be the matrix of ω under a given basis. ω being non-degenerate implies $\Omega \in \text{GL}(V)$. $\Omega^T = -\Omega$ and then $\det \Omega = (-1)^n \det \Omega$ is nonzero, which forces $m = 2n$.

Ω is a real normal matrix because $\Omega^T \Omega = \Omega^2 = \Omega \Omega^T$. Thus we can find an orthogonal matrix Q such that $Q^T \Omega Q = \text{diag}(A_1, \dots, A_n)$, where A_i is either a real number, or a 2×2 real matrix of the form $\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$.

Ω is both skew-symmetric and invertible, so each A_i should be a 2×2 matrix with zero diagonal elements. We obtain

$$Q^T \Omega Q = \begin{pmatrix} 0 & b_1 & & & \\ -b_1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & b_m \\ & & & -b_m & 0 \end{pmatrix}, b_i > 0.$$

Taking $P = Q \cdot \text{diag}(b_1^{-\frac{1}{2}}, b_1^{-\frac{1}{2}}, \dots, b_m^{-\frac{1}{2}}, b_m^{-\frac{1}{2}})$, we have

$$P^T \Omega P = \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix}.$$

In other words, we find a basis $v_1, u_1, \dots, v_m, u_m$ which meets the requirement. $\square$

Exercise 1.2. Prove that the product of two spheres $S^p \times S^q$ admits a symplectic structure if and only if $p = q = 1$ or $p = q = 2$.

Solution. As a priori, $p + q$ is even. For $p = q = 1$, $S^1 \times S^1$ is the torus, which is a symplectic manifold with a volume form. For $p = q = 2$, $S^2 \times S^2$ admits a symplectic structure as a product of the symplectic manifold S^2 .

$S^p \times S^q$ has Poincaré polynomial

$$(1 + t^p)(1 + t^q) = 1 + t^p + t^q + t^{p+q}.$$

If $p + q \geq 4$, then either $p = 2$ or $q = 2$ as $H^2(S^p \times S^q; \mathbb{R}) \neq 0$. Say $p = 2$. Then $q = 2, 4$. It remains to consider $S^2 \times S^4$. Notice that $H^2(S^2 \times S^4; \mathbb{R}) = \mathbb{R} \pi^*[\omega_0]$, where

ω_0 is a volume form on S^2 and π is the canonical projection on S^2 . Suppose ω is a symplectic structure on $S^2 \times S^4$. Then $[\omega] = a\pi^*[\omega_0]$ for some $a \neq 0$. It follows that $[\omega^3] = a^3\pi^*[\omega_0^3] = 0$, a contradiction, so $S^2 \times S^4$ has no symplectic structure. $\square$

Exercise 1.3. Prove that the canonical 1-form $\lambda_{\text{can}} = p_i dq^i$ for local coordinates $(q^1, \dots, q^n, p_1, \dots, p_n)$ of T^*Q is well-defined, i.e. the formula is compatible with coordinate changes.

Solution. Let $(x^1, \dots, x^n, y_1, \dots, y_n)$ be another local coordinates. By definition of T^*Q , the coordinate change is induced by a coordinate change $q^i = \varphi^i(x)$ on Q . Under coordinate change, we have

$$\varphi^* dq^i = d\varphi^i = \frac{\partial \varphi^i}{\partial x^j} dx^j = \sum_{j=1}^n \frac{\partial \varphi^i}{\partial x^j} dx^j.$$

Hence the coordinate change on T^*Q is given by

$$y_i = \frac{\partial \varphi^j}{\partial x^i} p_j.$$

As a consequence,

$$\varphi^*(p_i dq^i) = p_i \frac{\partial \varphi^i}{\partial x^j} dx^j = y_j dx^j.$$

A more invariant way to see this fact is noticing that

$$\lambda_{(q,p)}(X) = p_q(\pi_* X), X \in T_{(q,p)} T^*Q.$$

$\square$

Exercise 1.4. Given a smooth manifold Q and a diffeomorphism $f: Q \rightarrow Q$, prove that f lifts to a symplectomorphism $\varphi_f: T^*Q \rightarrow T^*Q$ such that $(\varphi_f)^* \lambda_{\text{can}} = \lambda_{\text{can}}$.

Solution. Define $\varphi_f(\alpha) = (f^* \alpha), \alpha \in T_q^*Q$, which is clearly a diffeomorphism. Under local coordinates, we have $\varphi_f: (q, p) \mapsto (q' = g(q), p')$, where g is the inverse of f . By definition,

$$p'_i d(q')^i = f^*(p_i dq^i) = p_i (df^i(q')) = p_i \frac{\partial f^i}{\partial (q')^j} d(q')^j,$$

hence $p'_i = p_j \frac{\partial f^j}{\partial (q')^i}$. It follows that

$$(\varphi_f) \lambda_{\text{can}} = p'_i d(q')^i = p_j \frac{\partial f^j}{\partial (q')^i} (dg^i(q)) = p_j \frac{\partial f^j}{\partial (q')^i} \frac{\partial g^i}{\partial q^k} dq^k = p_i dq^i = \lambda_{\text{can}}.$$

$\square$

Exercise 1.5. (i) Prove that ω_{FS} is well-defined on $\mathbb{CP}^n$, i.e. $(\omega_{\text{FS}}|_{U_i})|_{U_i \cap U_j} = (\omega_{\text{FS}}|_{U_j})|_{U_i \cap U_j}$.
(ii) Prove that ω_{FS} is a symplectic form.

(iii) Prove $\int_{\mathbb{CP}^1} \omega_{\text{FS}} = \pi$.

Solution. (i) Under the coordinates $(z_1, \dots, z_n)$ of $U_i \cong \mathbb{C}^n$,

$$\omega_{\text{FS}}|_{U_i} = \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k=1}^n |z_k|^2 + 1 \right).$$

Under coordinate change on $U_i \cap U_j$

$$(z_1, \dots, z_n) = \left(\frac{w_1}{w_i}, \dots, \frac{1}{w_i}, \dots, \frac{w_n}{w_i} \right),$$

we have

$$\begin{aligned} (\omega_{\text{FS}}|_{U_i})|_{U_i \cap U_j} &= \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k \neq j} \left| \frac{w_k}{w_i} \right|^2 + \left| \frac{1}{w_i} \right|^2 + 1 \right) \\ &= \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k=1}^n |w_k|^2 + 1 \right) - \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log |w_i|^2. \end{aligned}$$

$\partial \bar{\partial} \log |w_i|^2 = \partial \bar{\partial} (\log w_i + \log \bar{w}_i) = 0$, so

$$(\omega_{\text{FS}}|_{U_i})|_{U_i \cap U_j} = \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k=1}^n |w_k|^2 + 1 \right) = (\omega_{\text{FS}}|_{U_j})|_{U_i \cap U_j}.$$

(ii) This can be verified on each U_i separately. As $d = \partial + \bar{\partial}$, $\partial \bar{\partial} = -\bar{\partial} \partial$ and $\partial^2 = \bar{\partial}^2 = 0$,

$$d\omega_{\text{FS}}|_{U_i} = (\partial + \bar{\partial}) \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k=1}^n |z_k|^2 + 1 \right) = 0.$$

It remains to show that $\omega_{\text{FS}}|_{U_i}$ is a non-degenerate real 2-form. A straightforward computation yields

$$\begin{aligned} \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log \left(\sum_{k=1}^n |z_k|^2 + 1 \right) &= \frac{\sqrt{-1}}{2} \partial \frac{\sum z_k d\bar{z}_k}{1 + \sum |z_k|^2} \\ &= \frac{\sqrt{-1}}{2} \left(\frac{\sum dz_k \wedge d\bar{z}_k}{1 + \sum |z_k|^2} - \frac{\sum \bar{z}_k dz_k \wedge \sum z_l d\bar{z}_l}{(1 + \sum |z_k|^2)^2} \right) \\ &= \frac{\sqrt{-1}}{2} \left(\frac{\sum dz_k \wedge d\bar{z}_k}{1 + \sum |z_k|^2} - \frac{\sum \bar{z}_k z_l dz_k \wedge d\bar{z}_l}{(1 + \sum |z_k|^2)^2} \right). \end{aligned}$$

We see that $\overline{\omega_{\text{FS}}} = \omega_{\text{FS}}$, so ω_{FS} is real. Let X be a smooth real vector field on U_i and write $X = Y + \bar{Y}$ with Y being the $(1, 0)$ -component.

$$\begin{aligned} \omega_{\text{FS}}(X, JX) &= \omega_{\text{FS}}(Y + \bar{Y}, \sqrt{-1}Y - \sqrt{-1}\bar{Y}) = -2\sqrt{-1}\omega_{\text{FS}}(Y, \bar{Y}) \\ &= \left(\frac{\sum |Y_k|^2}{1 + \sum |z_k|^2} + \frac{\sum \bar{z}_k z_l Y_k \bar{Y}_l}{(1 + \sum |z_k|^2)^2} \right) = \frac{(1 + |z|^2) |Y|^2 - \langle \bar{z}, Y \rangle}{1 + |z|^2} \geq 0, \end{aligned}$$

where $\langle -, - \rangle$ is the standard Hermitian form $\mathbb{C}^n$. This shows that ω_{FS} is indeed a symplectic form.

(iii) $\mathbb{CP}^1 \setminus U_0$ is a measure zero set, hence

$$\begin{aligned} \int_{\mathbb{CP}^1} \omega_{\text{FS}} &= \int_{\mathbb{C}} \frac{\sqrt{-1}}{2} \partial \bar{\partial} \log(|z|^2 + 1) \\ &= \int_{\mathbb{C}} \frac{\sqrt{-1}}{2} \frac{dz \wedge d\bar{z}}{(|z|^2 + 1)^2} = \int_{\mathbb{R}^2} \frac{dx \wedge dy}{(x^2 + y^2 + 1)^2} \\ &= 2\pi \int_0^\infty \frac{r dr}{r^2 + 1} = \pi. \end{aligned}$$

□

Exercise 1.6. For an invertible skew-symmetric matrix A , prove that A and $(AA^T)^{-\frac{1}{2}}$ commute.

Solution. Let $B = (AA^T)^{-\frac{1}{2}}$. Then

$$(ABA^{-1})^2 = AB^2A^{-1} = AA^{-1}A^{-1} = -A^{-2} = (AA^T)^{-1}.$$

By the uniqueness of B , $B = ABA^{-1}$, which implies $AB = BA$.

□

Exercise 1.7. For any symplectic manifold (M, ω) , prove that the set of all ω -compatible almost complex structures $\mathcal{J}(M, \omega)$ is path-connected.

Solution. Take $J_0, J_1 \in \mathcal{J}(M, \omega)$ and define $g_i = \omega(-, J_i -)$. Then g_0, g_1 are both metrics on M . Now let $g_t = (1-t)g_0 + tg_1$. For each $t \in [0, 1]$, there exists $J_t \in \mathcal{J}(M, \omega)$ such that $g_t = \omega(-, J_t -)$. By the construction of J_t , $t \mapsto J_t$ defines a path in $\mathcal{J}(M, \omega)$ joining J_0, J_1 .

□

Exercise 1.8. If we define $\overline{Z}^{2n}(r)$ by

$$\{(x_1, x_2) \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq r^2\} \times \mathbb{R}^{2n-2},$$

then prove that $B^{2n}(R)$ symplectically embeds into $\overline{Z}^{2n}(r)$ for any $r > 0$.

Solution. Consider $\varphi(x_1, y_1, x_2, y_2) = (\frac{r}{R}x_1, \frac{R}{r}y_1, \frac{r}{R}x_2, \frac{R}{r}y_2)$. Then $\varphi^*\omega_{\text{std}} = \omega_{\text{std}}$.

□

Exercise 1.9. For $(\mathbb{R}^{2n}, \omega_{\text{std}})$, consider $H(x_1, y_1, \dots, x_n, y_n) = \sum_{i=1}^n (x_i^2 + y_i^2)$. Compute the resulting Hamiltonian flow φ_H^t .

Solution. Let $X_H = \sum a_i \partial_{x_i} + b_i \partial_{y_i}$ be the Hamiltonian vector field generated by H . Then

$$\iota_{X_H} \omega_{\text{std}} = \sum_{i=1}^n (a_i dy_i - b_i dx_i).$$

Since $dH = \sum (2x_i dx_i + 2y_i dy_i)$, we have $a_i = -2y_i, b_i = 2x_i$. For $(x, y) = (p, q)$, $\varphi_H^t((p, q)) = (x(t), y(t))$ is the integral curve along X_H , so we obtain an ODE

$$\begin{cases} \dot{x}(t) = -2y(t) \\ \dot{y}(t) = 2x(t) \\ x(0) = q, y(0) = p. \end{cases}$$

We see that $\varphi_H^t(q, p) = (q \cos(2t) - p \sin(2t), q \sin(2t) + p \cos(2t))$. $\square$

Exercise 1.10. On the torus $(\mathbb{T}^2, \omega_{\text{area}})$, find a symplectomorphism φ such that it is not a Hamiltonian diffeomorphism. Moreover, justify your answer.

Solution. Let x, y be the local angle coordinates of $\mathbb{T}^2$. By Moser's isotopy technique, we may assume $\omega_{\text{area}} = dx \wedge dy$. Consider $\varphi(x, y) = (x + \frac{1}{2}, y + \frac{1}{2})$, which is a symplectomorphism.

Suppose ψ is a Hamiltonian diffeomorphism. Then we can find an isotopy ψ_t generated by a $\mathbb{Z}^2$ -period Hamiltonian $H_t(x, y)$. Every diffeomorphism on $\mathbb{T}^2$ can be written as $(x, y) \mapsto (x + \alpha(x, y), y + \beta(x, y))$ where α, β are $\mathbb{Z}^2$ -period functions on $\mathbb{R}^2$, by considering its lifting to $\mathbb{R}^2$. Then we write $\psi_t(x, y) = (x + \alpha_t(x, y), y + \beta_t(x, y))$ as mentioned above. By definition, $X_H^t(x, y) = \frac{d\alpha_t}{dt} \partial_x + \frac{d\beta_t}{dt} \partial_y$. On the other hand,

$$\frac{\partial H_t}{\partial x} dx + \frac{\partial H_t}{\partial y} dy = dH = -\iota_{X_H^t}(dx \wedge dy) = \frac{d\beta_t}{dt} dx - \frac{d\alpha_t}{dt} dy.$$

Now consider integral

$$\int_0^1 \alpha_t dx \wedge dy.$$

Differentiate with respect to t to obtain

$$\frac{d}{dt} \int_{\mathbb{T}^2} \alpha_t dx \wedge dy = \int_{\mathbb{T}^2} \frac{d\alpha_t}{dt} dx \wedge dy = - \int_{\mathbb{T}^2} \frac{\partial H_t}{\partial y} dx \wedge dy = \int_{\mathbb{T}^2} d(H_t dx) = 0.$$

It follows that

$$\int_{\mathbb{T}^2} \alpha_1 dx \wedge dy = \int_{\mathbb{T}^2} \alpha_0 dx \wedge dy = 0,$$

as $\psi_0 = \mathbb{1}$. However, for $\psi = \varphi$, we have

$$\int_{\mathbb{T}^2} \alpha_1 dx \wedge dy = \int_{\mathbb{T}^2} \frac{1}{2} dx \wedge dy = \frac{1}{2}.$$

Hence φ is not Hamiltonian, otherwise we get a contradiction. $\square$

2 HOMEWORK 2

Exercise 2.1. Let V be a finite dimensional $\mathbb{R}$ -vector space and ω be a symplectic form. Let W be a isotropic subspace of V , i.e. $\omega|_W = 0$. Prove that $\dim W \leq \frac{1}{2} \dim V$.

Solution. Consider $f: V \rightarrow V^*$ by setting $f(v)(w) = \omega(v, w)$. Since ω is non-degenerate, f is an isomorphism. Under f , $W^\omega = \{v \in V \mid \omega(v, w) = 0\}$ is identified with $W^\perp \subset V^*$. It follows that $\dim W + \dim W^\omega = \dim V$. Since W is isotropic, $W \subset W^\omega$ and the conclusion follows. $\square$

Exercise 2.2. For a Riemannian surface (M, g) with the metric $g = (g_{ij})$, consider a function $L: TM \rightarrow \mathbb{R}$ locally defined by

$$L(q_1, q_2, v_1, v_2) = \frac{1}{2} \sum_{i,j} g_{ij} v_i v_j.$$

Compute the Legendre transform of $L(q, \cdot)$.

Solution. Fix $q \in M$, then

$$\mathcal{L}: T_q M \rightarrow T_q^* M, (v_1, v_2) \mapsto (g_{11}v_1 + g_{12}v_2, g_{12}v_1 + g_{22}v_2).$$

Hence the inverse map is

$$\mathcal{L}^{-1}: T_q^* M \rightarrow T_q M, (p_1, p_2) \mapsto (g^{11}p_1 + g^{12}p_2, g^{12}p_1 + g^{22}p_2).$$

The the Legendre transform of $L(q, \cdot)$ is

$$H(q, p) = \sum_{i,j} g^{ij} p_i p_j - \frac{1}{2} \sum_{i,j} g_{ij} \sum_k g^{ik} p_k \sum_l g^{jl} p_l = \frac{1}{2} \sum_{i,j} g^{ij} p_i p_j.$$

$\square$

Exercise 2.3. Prove that a non-degenerate 2-form ω on a manifold M is closed if and only if the Poisson bracket satisfies the following Jacobi identity,

$$\{F, \{G, H\}\} + \{G, \{H, F\}\} + \{H, \{F, G\}\} = 0$$

for any $F, G, H \in C^\infty(M)$.

Solution. By definition, we have

$$\begin{aligned} \{F, \{G, H\}\} &= X_F \omega(X_G, X_H) \\ &= (\mathcal{L}_{X_F} \omega)(X_G, X_H) + \omega(\mathcal{L}_{X_F} X_G, X_H) - \omega(X_G, \mathcal{L}_{X_F} X_H) \\ &= (\iota_{X_F} d\omega + d\iota_{X_F} \omega)(X_G, X_H) + \omega([X_F, X_G], X_H) + \omega(X_G, [X_F, X_H]) \\ &= \omega([X_F, X_G], X_H) + \omega(X_G, [X_F, X_H]). \end{aligned}$$

We can derive the similar formula for $\{G, \{H, F\}\}, \{H, \{F, G\}\}$, which implies

$$\{F, \{G, H\}\} + \{G, \{H, F\}\} + \{H, \{F, G\}\} = 2(d\omega)(X_F, X_G, X_H).$$

If ω is closed, the Jacobi identity holds. Now we want to show that the Jacobi identity implies $d\omega = 0$. Note that this is a local condition. A vector field X is Hamiltonian if and only if $\iota_X\omega$ is exact, while every 1-form is locally exact. It follows that for every $v \in T_x^*M$, we can find a Hamiltonian vector field V such that $V_x = v$, which implies $(d\omega)_x = 0$. $\square$

Exercise 2.4. Let L be an n -dimensional submanifold of a symplectic manifold (M, ω) with $\dim M = 2n$. Consider the following subspace of $C^\infty(M)$,

$$I_L = \{F \in C^\infty(M) \mid F|_L = 0\}.$$

Prove that L is Lagrangian submanifold if and only if the Poisson bracket is closed in I_L , that is, $\{F, G\}|_L = 0$ for any $F, G \in I_L$.

Solution. First assume that L is Lagrangian. Since $F|_L, G|_L = 0$, both dF, dG vanish on T_xL for every $x \in L$. It follows that $X_F, X_G \in (T_xL)^\omega = T_xL$ for every $x \in L$. Therefore,

$$\{F, G\}|_L = \omega(X_F, X_G)|_L = 0.$$

Now assume that Poisson bracket is closed in I_L , and we want to show that L is Lagrangian. By contradiction, suppose there exists $x \in M$ such that $\omega(u, v) \neq 0$ for some $u, v \in T_xL$. We can take $G \in I_L$ such that $\omega_x(X_G, v) = -dG_x(v) \neq 0$, which is always possible if we work in a local coordinate. In particular, $X_G \notin T_xL$. In the same way, we can construct $F \in I_L$ such that $\omega_x(X_F, X_G) = -dF(X_G) \neq 0$. However, $\{F, G\}(x) \neq 0$ violates our hypothesis, so L must be Lagrangian. $\square$

Exercise 2.5. Let $\mathfrak{so}(n)$ be the Lie algebra of $\mathrm{SO}(n)$. Compute the Lie bracket of $\mathfrak{so}(n)$.

Solution. $\mathfrak{gl}(n, \mathbb{R}) = M(n, \mathbb{R})$ is the Lie algebra of $\mathrm{GL}^+(n, \mathbb{R})$, whose Lie bracket is $[x, y] = xy - yx$. Let $F: \mathrm{GL}^+(n, \mathbb{R}) \rightarrow M(n, \mathbb{R}), A \mapsto AA^T$. $\mathrm{SO}(n) = F^{-1}(\mathbf{1})$ is a Lie subgroup of $\mathrm{SO}(n)$, so $\mathfrak{so}(n) = \ker(dF)_1 = \{x \in \mathfrak{gl}(n, \mathbb{R}) \mid x + x^T = 0\}$ is a Lie subalgebra of $\mathfrak{gl}(n, \mathbb{R})$. In particular, the Lie bracket of $\mathfrak{so}(n)$ is given by $[x, y] = xy - yx$. We can write out an element of $\mathfrak{so}(n)$ as

$$x = \begin{pmatrix} 0 & -a_3 & -a_2 \\ a_3 & 0 & -a_1 \\ a_2 & a_1 & 0 \end{pmatrix}$$

to identify x with $(a_1, a_2, a_3) \in \mathbb{R}^3$. Under the identification, the Lie bracket on $\mathbb{R}^3$ is given by the cross product. $\square$

Exercise 2.6. Consider the $\mathbb{T}^2$ -action on $(\mathbb{CP}^2, \omega_{\mathrm{FS}})$ defined by

$$(t_1, t_2) \cdot [z_0, z_1, z_2] = [z_0, e^{\sqrt{-1}t_1}z_1, e^{\sqrt{-1}t_2}z_2],$$

where $(t_1, t_2) \in (\mathbb{R}/2\pi\mathbb{Z})^2 = \mathbb{T}^2$ and $[z_0, z_1, z_2] \in \mathbb{CP}^2$. Prove that the moment map of the action is

$$\mu([z_0, z_1, z_2]) = \frac{1}{2} \left(\frac{|z_1|^2}{\sum |z_i|^2}, \frac{|z_2|^2}{\sum |z_i|^2} \right).$$

Solution. $\mathbb{T}^2$ has Lie algebra $\mathfrak{g} = \mathbb{R}^2$ and the corresponding exponential map is $(t, v) \mapsto tv \in (\mathbb{R}/2\pi\mathbb{Z})^2$. Under local coordintae $(x, y) \mapsto [1, x, y]$ we have

$$\exp(tv) \cdot (x, y) = (e^{\sqrt{-1}tv_1}x, e^{\sqrt{-1}tv_2}y),$$

for $v = (v_1, v_2) \in \mathfrak{g}$. Therefore,

$$X_v = \frac{d}{dt} \Big|_{t=0} \exp(tv) \cdot (x, y) = (\sqrt{-1}v_1x, \sqrt{-1}v_2y).$$

Then

$$X_v = \sqrt{-1}v_1 (x\partial_x - \bar{x}\partial_{\bar{x}}) + \sqrt{-1}v_2 (y\partial_y - \bar{y}\partial_{\bar{y}}).$$

Locally ω_{FS} writes

$$\frac{\sqrt{-1}}{2} \frac{dx \wedge d\bar{x} + dy \wedge d\bar{y}}{1 + |x|^2 + |y|^2} - \frac{\sqrt{-1}}{2} \frac{(\bar{x}dx + \bar{y}dy) \wedge (xd\bar{x} + yd\bar{y})}{(1 + |x|^2 + |y|^2)^2},$$

so we obtain

$$\begin{aligned} -\iota_{X_v}\omega_{\text{FS}} &= \frac{v_1 d|x|^2 + v_2 d|y|^2}{2(1 + |x|^2 + |y|^2)} - \frac{(v_1 |x|^2 + v_2 |y|^2)(d|x|^2 + d|y|^2)}{2(1 + |x|^2 + |y|^2)^2} \\ &= \frac{1}{2}d \left(v_1 \frac{|x|^2}{1 + |x|^2 + |y|^2} + v_2 \frac{|y|^2}{1 + |x|^2 + |y|^2} \right). \end{aligned}$$

Similar computation under other local coordinates shows that the moment map is given by

$$\mu([z_0, z_1, z_2]) = \frac{1}{2} \left(\frac{|z_1|^2}{\sum |z_i|^2}, \frac{|z_2|^2}{\sum |z_i|^2} \right).$$

□

Exercise 2.7. Let $M = \{([a, b], [x, y, z]) \in \mathbb{CP}^1 \times \mathbb{CP}^2 \mid ay = bx\}$ be a symplectic form $\omega_{\text{prod}}|_M$ and $\mathbb{T}^2$ -action on $(M, \omega_{\text{prod}}|_M)$ be

$$(t_1, t_2) \cdot ([a, b], [x, y, z]) = \left([e^{\sqrt{-1}t_1}a, b], [e^{\sqrt{-1}t_1}x, y, e^{\sqrt{-1}t_2}z] \right).$$

Compute the moment map μ of the action and find the fixed points of this action.

Solution. $\mathbb{T}^2$ has Lie algebra $\mathfrak{g} = \mathbb{R}^2$ and the corresponding exponential map is $(t, v) \mapsto tv \in (\mathbb{R}/2\pi\mathbb{Z})^2$. Under local coordintae $(u, z) \mapsto ([u, 1], [u, 1, z])$ we have

$$\exp(tv) \cdot (u, z) = (e^{\sqrt{-1}tv_1}u, e^{\sqrt{-1}tv_2}z),$$

for $v = (v_1, v_2) \in \mathfrak{g}$. Therefore,

$$X_v = \frac{d}{dt} \Big|_{t=0} \exp(tv) \cdot (u, z) = (\sqrt{-1}v_1u, \sqrt{-1}v_2z).$$

Then

$$X_v = \sqrt{-1}v_1 (u\partial_u - \bar{u}\partial_{\bar{u}}) + \sqrt{-1}v_2 (z\partial_z - \bar{z}\partial_{\bar{z}}).$$

Locally $\omega_{\text{prod}}|_M$ writes

$$\frac{\sqrt{-1}}{2} \frac{du \wedge d\bar{u}}{(1 + |u|^2)^2} + \frac{\sqrt{-1}}{2} \frac{du \wedge d\bar{u} + dz \wedge d\bar{z}}{1 + |u|^2 + |z|^2} - \frac{\sqrt{-1}}{2} \frac{(\bar{u}du + \bar{z}dz) \wedge (ud\bar{u} + zd\bar{z})}{(1 + |u|^2 + |z|^2)^2},$$

so we obtain

$$\begin{aligned} -\iota_{X_v}\omega_{\text{FS}} &= \frac{v_1 d|u|^2}{2(1 + |u|^2)^2} + \frac{v_1 d|u|^2 + v_2 d|z|^2}{2(1 + |u|^2 + |z|^2)} - \frac{(v_1 |u|^2 + v_2 |z|^2)(d|u|^2 + d|z|^2)}{2(1 + |u|^2 + |z|^2)^2} \\ &= \frac{1}{2}d \left(v_1 \frac{1}{1 + |u|^2} + v_1 \frac{|u|^2}{1 + |u|^2 + |z|^2} + v_2 \frac{|z|^2}{1 + |u|^2 + |z|^2} \right). \end{aligned}$$

Similar computation under other local coordinates shows that the moment map is given by

$$\mu([z_0, z_1, z_2]) = \frac{1}{2} \left(\frac{|b|^2}{|a|^2 + |b|^2} + \frac{|x|^2}{|x|^2 + |y|^2 + |z|^2}, \frac{|z|^2}{|x|^2 + |y|^2 + |z|^2} \right).$$

□

Suppose $([a_0, b_0], [x_0, y_0, z_0])$ is a fixed point of this action. Then

$$\left(\left[e^{\sqrt{-1}t_1} a_0, b_0 \right], \left[e^{\sqrt{-1}t_1} x_0, y_0, e^{\sqrt{-1}t_2} z_0 \right] \right) = ([a_0, b_0], [x_0, y_0, z_0])$$

for all $t_1, t_2 \in \mathbb{R}$. From the first component we have either $a_0 = 0$ or $b_0 = 0$, and from the second component we have $[x_0, y_0, z_0] = [1, 0, 0], [0, 1, 0]$ or $[0, 0, 1]$. By considering the possible combinations of components, we see that the fixed points are

$$([1, 0], [1, 0, 0]), ([1, 0], [0, 0, 1]), ([0, 1], [0, 1, 0]), ([0, 1], [0, 0, 1]).$$

Exercise 2.8. Let $H, H_2: T^*(\mathbb{R}^3 \setminus \{0\}) \rightarrow \mathbb{R}$ be

$$\begin{aligned} H(q, p) &= \frac{1}{2} |p|^2 - \frac{1}{|q|}, \\ H_2(q, p) &= q_1 p_2 - q_2 p_1. \end{aligned}$$

Recall that the Runge–Lenz vector is given by $((V_{\text{RL}})_1, (V_{\text{RL}})_2, (V_{\text{RL}})_3) = p \times (q \times p) - \frac{q}{|q|}$. Prove that $T^*(\mathbb{R}^3 \setminus \{0\}), \omega_{\text{std}}, H)$ is an integrable system by considering $(H, H_2, (V_{\text{RL}})_3)$.

Solution. We have

$$\begin{aligned} dH &= \sum_{i=1}^3 p_i dp_i + \frac{1}{|q|^3} \sum_{i=1}^3 q_i dq_i, \\ dH_2 &= p_2 dq_1 + q_1 dp_2 - p_1 dq_2 - q_2 dp_1. \end{aligned}$$

Since $p \times (q \times p) = |p|^2 q - (p \cdot q)p$,

$$\begin{aligned} d((V_{\text{RL}})_3) &= d\left(|p|^2 q_3 - (p \cdot q)p_3 - \frac{q_3}{|q|}\right) \\ &= 2q_3 \sum_{i=1}^3 p_i dp_i + |p|^2 dq_3 - p_3 \sum_{i=1}^3 (p_i dq_i + q_i dp_i) - (p \cdot q)dp_3 - \frac{1}{|q|} dq_3 + \frac{1}{|q|^3} \sum_{i=1}^3 q_3 q_i dq_i. \end{aligned}$$

The three differentials are linearly independent on $T^*(\mathbb{R}^3 \setminus \{0\})$. Since $\omega_{\text{std}} = \sum dq_i \wedge dp_i$, the corresponding Hamiltonian vector fields are

$$\begin{aligned} X_H &= -\sum_{i=1}^3 p_i \frac{\partial}{\partial q_i} + \frac{1}{|q|^3} \sum_{i=1}^3 q_i \frac{\partial}{\partial p_i}, \\ X_{H_2} &= q_2 \frac{\partial}{\partial q_1} + p_2 \frac{\partial}{\partial p_1} - q_1 \frac{\partial}{\partial q_2} - p_1 \frac{\partial}{\partial p_2}, \\ X_{(V_{\text{RL}})_3} &= -\sum_{i=1}^3 (2p_i q_3 - p_3 q_i) \frac{\partial}{\partial q_i} + \sum_{i=1}^3 \left(\frac{1}{|q|^3} q_3 q_i - p_i p_3 \right) \frac{\partial}{\partial p_i} \\ &\quad - (p \cdot q) \frac{\partial}{\partial q_3} + \left(|p|^2 - \frac{1}{|q|} \right) \frac{\partial}{\partial p_3}. \end{aligned}$$

Therefore,

$$\begin{aligned} \{H, H_2\} &= -p_1 p_2 - \frac{q_1 q_2}{|q|^3} - p_2 \cdot (-p_1) - \frac{(-q_1)q_2}{|q|^3} = 0, \\ \{H, (V_{\text{RL}})_3\} &= \sum_{i=1}^3 p_i \left(p_i p_3 - \frac{q_3 q_i}{|q|^3} \right) + \frac{1}{|q|^3} \sum_{i=1}^3 q_i (2p_i q_3 - p_3 q_i) \\ &\quad - \left(|p|^2 - \frac{1}{|q|} \right) p_3 + \frac{q_3}{|q|^3} (p \cdot q) = 0, \\ \{H_2, (V_{\text{RL}})_3\} &= q_2 \left(\frac{1}{|q|^3} q_3 q_1 - p_1 p_3 \right) + p_2 (2p_1 q_3 - p_3 q_1) \\ &\quad - q_1 \left(\frac{1}{|q|^3} q_3 q_2 - p_2 p_3 \right) - p_1 (2p_2 q_3 - p_3 q_2) = 0. \end{aligned}$$

We see that $T^*(\mathbb{R}^3 \setminus \{0\}), \omega_{\text{std}}, H, H_2, (V_{\text{RL}})_3$ is an integrable system. $\square$

Exercise 2.9. For an effective Hamiltonian $\mathbb{T}^n$ -action on a symplectic manifold (M, ω) with $\dim M = 2n$, let $\mu = (\mu_1, \dots, \mu_n): M \rightarrow \mathbb{R}^n$ be the moment map of the action. Prove that (M, ω, μ_i) is an integrable system for $i = 1, \dots, n$.

Proof. Let $\mathfrak{g} = \mathbb{R}^n$ be the Lie algebra of $\mathbb{T}^n$. $v_i = \partial_{\theta_i}$ form a basis for $\mathfrak{g}$, with $[v_i, v_j] = 0$. Then $d\mu_i = -\iota_{X_i} \omega$, where

$$X_i(p) = \left. \frac{d}{dt} \right|_{t=0} \exp(tv_i) \cdot p,$$

so $\{\mu_i, \mu_j\} = \omega(X_i, X_j) = \mu([v_i, v_j]) = 0$.

It remains to show that $d\mu_1, \dots, d\mu_n$ are linearly independent on an open dense subset of M . Let A be this set, and it is open by the following observation: $p \in A$ if and only if $d\mu_p: T_p M \rightarrow \mathfrak{g}^*$ is surjective.

For $p \in M$, let $X_p: \mathfrak{g} \rightarrow T_p M, v \mapsto X_v(p)$, where X_v is the Hamiltonian vector field generated by $H_v(x) = \langle \mu(x), v \rangle$. Since $\iota_{X_v} \omega = -dH_v = -\langle d\mu_p, v \rangle$, $\ker X_p$ is the annihilator of $\text{im } d\mu_p \subset \mathfrak{g}^*$. We see that $p \in A$ if and only if the stabilizer $G_p \leq G$ is discrete. Then $q \in M \setminus A$ has stabilizer whose dimension is at least one. Note that a stabilizer is closed.

Let $K \leq G$ be a connected closed subgroup of dimension. Then there is a nonzero $w \in \mathfrak{g}_K$ generating a dense subgroup of K . It follows that $q \in M \setminus A$ fixed by K are the same as the critical points of H_w . By Atiyah–Guillemin–Sternberg theory, H_w is either constant or a Morse function. As the torus action is effective, H_w is not constant for $w \neq 0$. Recall that compact Lie subgroups of torus are countable, so A is the intersection of countable open dense set, and is therefore dense. $\square$

Exercise 2.10. For $\mathbb{T}^2 = (\mathbb{R}/2\pi\mathbb{Z})^2$ with angular coordinates φ, θ and $0 < \varepsilon < \frac{1}{2}$, let $g = (1 + \varepsilon \sin \varphi)^2 d\varphi \otimes d\varphi + d\theta \otimes d\theta$ be a metric on $\mathbb{T}^2$. Let $L: T\mathbb{T}^2 \rightarrow \mathbb{R}$ be as in exercise 2.2. Compute the Legendre transform H of L and prove that, for sufficiently small ε , $(T^*\mathbb{T}^2, d\lambda_{\text{can}}, H)$ is an integrable system together with a Hamiltonian H_2 . Compute the Hamiltonian flow of H and H_2 .

Solution. For sufficiently small $\varepsilon > 0$, g is indeed a metric, so we can carry out the Legendre transform. By Exercise 2.2,

$$H(\varphi, \theta, u, v) = \frac{1}{2}(1 + \varepsilon \sin \varphi)^{-2}u^2 + \frac{1}{2}v^2.$$

Note that $H_2(\varphi, \theta, u, v) = v$ is a well-defined smooth function (since $T^*\mathbb{T}^2 = \mathbb{T}^2 \times \mathbb{R}^2$). As $dH_2(X_H) = 0$ and $dH, dH_2 = dy$ is linearly independent, $(T^*\mathbb{T}^2, d\lambda_{\text{can}}, H, H_2)$ is an integrable system. The Hamiltonian vector field is

$$X_H = (1 + \varepsilon \sin \varphi)^{-2}u \frac{\partial}{\partial \varphi} + \varepsilon(1 + \varepsilon \sin \varphi)^{-3}u^2 \cos \varphi \frac{\partial}{\partial u} + v \frac{\partial}{\partial \theta},$$

which determines an ODE

$$\begin{cases} \dot{\varphi} = (1 + \varepsilon \sin \varphi)^{-2}u, \\ \dot{u} = \varepsilon(1 + \varepsilon \sin \varphi)^{-3}u^2 \cos \varphi, \\ \dot{\theta} = v, \\ \dot{v} = 0. \end{cases}$$

Given initial conditions, the flow is determined by

$$\begin{cases} u_0(1 + \varepsilon \sin \varphi) = u(1 + \varepsilon \sin \varphi_0), \\ \theta = \theta_0 + v_0 t, \\ v = v_0. \end{cases}$$

The Hamiltonian flow generated by H_2 is translation in θ . $\square$

3 HOMEWORK 3

Exercise 3.1. Let $(M, \xi = \ker \alpha)$ be a contact manifold and $\beta = e^f \alpha$ for $f \in C^\infty(M)$. Compute the Reeb vector field of β in terms of the Reeb vector field R_α of α .

Solution. It suffices to compute R_β pointwise. Take $p \in M$. Since $d\alpha|_\xi$ is non-degenerate, we may pick a basis X_i, Y_i of ξ_p such that $\xi_p(X_i, Y_i) = \delta_{ij}$. We can write

$$R_\beta(p) = \sum_{i=1}^n (a_i X_i + b_i Y_i) + c R_\alpha.$$

$1 = \beta(R_\beta) = e^f \alpha(R_\beta) = c e^f$, so $c = e^{-f}$. Direct computation yields $d\beta = e^f df \wedge \alpha + e^f d\alpha$, so

$$\begin{aligned} 0 = d\beta(R_\beta, X_i) &= e^f (R_\beta f) \alpha(X_i) - e^f (X_i f) \alpha(R_\beta) + e^f d\alpha(R_\beta, X_i) \\ &= b_i e^f + e^f (X_i f) e^{-f} = b_i e^f + (X_i f). \end{aligned}$$

Similarly, we obtain

$$0 = d\beta(R_\beta, Y_i) = a_i e^f - (Y_i f).$$

We see that

$$R_\beta(p) = e^{-f} \sum_{i=1}^n ((Y_i f) X_i - (X_i f) Y_i) + e^{-f} R_\alpha.$$

□

Exercise 3.2. For an exact symplectic manifold $(W, d\lambda)$, consider the symplectization of the contactization $(CW = W \times S^1, \ker(\lambda + d\theta))$ as follows:

$$S_+(CW) := (\mathbb{R}_{>0} \times CW, d(r(\lambda + d\theta))).$$

Prove that $S_+(CW)$ is symplectomorphic to the product $(W \times T_{>0}^* S^1, \pi_1^* d\lambda + \pi_2^* (dr \wedge d\theta))$ where $(r, \theta) \in \mathbb{R}_{>0} \times S^1 = T_{>0}^* S^1$ and π_1, π_2 are projections to W and $T_{>0}^* S^1$, respectively.

Solution. Since $d\lambda$ is non-degenerate, we can find $X \in \Gamma(TW)$ such that $\iota_X d\lambda = \lambda$. Let $\{\varphi_t\}$ be the flow generated by X , then

$$\frac{d}{dt}(\varphi_t^* \lambda) = \varphi_t^* \mathcal{L}_X \lambda = \phi_t^* (\iota_X d\lambda + d(\lambda(X))) = \varphi_t^* \lambda.$$

Since $\varphi_0 = \mathbb{1}$, $\varphi_t^* \lambda = e^t \lambda$. Let $f: S_+(CW) \rightarrow W \times T_{>0}^* S^1, (r, x, \theta) \mapsto (\varphi_{\log r}(x), r, \theta)$. Direct computation yields

$$\begin{aligned} F^*(d\lambda + dr \wedge d\theta) &= d(\varphi_{\log r}^* \lambda) + dr \wedge d\theta \\ &= d(r\lambda) + dr \wedge d\theta = d(r(\lambda + d\theta)). \end{aligned}$$

□

Exercise 3.3. For a submanifold N of a Riemannian manifold (Q, g) , prove that the unit conormal bundle Λ_N is a Legendrian submanifold of $(S_g^*M, \lambda_{\text{can}}|_{S_g^*M})$.

Solution. Suppose $\dim N = k$ and $\dim Q = n$. Then $\dim \Lambda_N = k + (n - k - 1) = n - 1$. Let $q_1, \dots, q_n$ be local coordinates on Q such that N is defined via q_i for $k + 1 \leq i \leq n$. Then $\bar{\Lambda}_N$ admits local coordinates $q_1, \dots, q_k, p_{k+1}, \dots, p_n$. As a consequence, $\lambda_{\text{can}} = 0$ on $\bar{\Lambda}_N$. Since $\Lambda_N = \bar{\Lambda}_N \cap S_g^*M$ is a submanifold of $\bar{\Lambda}_N$, we see that $\Lambda_N = 0$ on Λ_N . $\square$

Exercise 3.4. Denotes the coordinates on $S^1 \times (\mathbb{R}^2 \setminus \{(0, 0)\})$ by θ, x, y and consider the 1-form

$$\alpha = \sin\left(\frac{x^2 + y^2}{2}\right) \frac{ydx - xdy}{x^2 + y^2} - \cos\left(\frac{x^2 + y^2}{2}\right) d\theta.$$

Show that it is a contact form with $\alpha \wedge d\alpha = d\theta \wedge dx \wedge dy$. Show that the rays $\theta = \text{const}, x/y = \text{const}$ and the circles $\theta = \text{const}, (x^2 + y^2)/2 = k\pi$ (with $k \in \mathbb{Z}_{>0}$) are Legendrian.

Solution. We have

$$\begin{aligned} d\alpha &= \cos\left(\frac{x^2 + y^2}{2}\right) (x dx + y dy) \wedge \frac{y dx - x dy}{x^2 + y^2} + \sin\left(\frac{x^2 + y^2}{2}\right) (x dx + y dy) \wedge d\theta \\ &= \cos\left(\frac{x^2 + y^2}{2}\right) dx \wedge dy + \sin\left(\frac{x^2 + y^2}{2}\right) (x dx + y dy) \wedge d\theta, \end{aligned}$$

so

$$\alpha \wedge d\alpha = \sin^2\left(\frac{x^2 + y^2}{2}\right) dx \wedge dy \wedge d\theta - \cos^2\left(\frac{x^2 + y^2}{2}\right) d\theta \wedge dx \wedge dy = d\theta \wedge dx \wedge dy.$$

It remains to show that α vanishes on the two 1-dimensional submanifolds. Let

$$\begin{aligned} S_1 &= \left\{ (\theta, x, y) \mid \theta = \text{const}, \frac{x}{y} = \text{const} \right\}, \\ S_2 &= \left\{ (\theta, x, y) \mid \theta = \text{const}, \frac{x^2 + y^2}{2} = k\pi \right\}. \end{aligned}$$

On S_1 , we have $d\theta = 0$ and $x dy - y dx = 0$, so $\alpha|_{S_1} = 0$. On S_2 , we have $d\theta = 0$ and $\sin \frac{x^2 + y^2}{2} = 0$, so $\alpha|_{S_2} = 0$. $\square$

Exercise 3.5. Let $(M, \xi = \ker \alpha)$ be a contact manifold and let $SM_\alpha := (\mathbb{R} \times M, d(e^r \alpha))$ be the symplectization of M associated with the contact form α .

- (i) Let $f : M \rightarrow \mathbb{R}$ be a smooth positive function. Prove that SM_α and $SM_{f\alpha}$ are symplectomorphic via $(r, x) \mapsto (r - \ln(f(x)), x)$.
- (ii) Prove that $L \subset M$ is a Legendrian submanifold of (M, ξ) if and only if $\mathbb{R} \times L$ is a Lagrangian submanifold of the symplectization SM_α .
- (iii) Prove that $\psi : M \rightarrow M$ is a contactomorphism of (M, ξ) with $\psi^* \alpha = e^{\kappa(\psi)} \alpha$ if and only if the map $\tilde{\psi}(r, x) := (r - \kappa(\psi)(x), \psi(x))$ is a symplectomorphism of SM_α .

Solution. (i) This is true because $e^{r-\ln f(x)}f\alpha = e^r\alpha$.

(ii) $d(e^r\alpha) = e^rd\alpha + e^rdr \wedge \alpha$. For every $X \in \Gamma(TL)$,

$$d(e^r\alpha)(\partial_r, X) = e^rd\alpha(\partial_r, X) + e^rdr \wedge \alpha(\partial_r, X) = e^r\alpha(X),$$

so $\alpha|_L = 0$ if and only if $d(e^r\alpha)|_{\mathbb{R} \times L} = 0$. On the other hand, $\dim L = \frac{1}{2}(\dim M - 1)$ if and only if $\dim L + 1 = \dim \mathbb{R} \times L = \frac{1}{2} \dim SM_\alpha = \frac{1}{2}(\dim M + 1)$.

(iii)

Suppose ψ is a contactomorphism with $\psi^*\alpha = e^{\kappa(\psi)}\alpha$. Then

$$\tilde{\psi}^*d(e^r\alpha) = d(e^{r-\kappa(\psi)}\psi^*\alpha) = d(e^r\alpha).$$

$(r, x) \mapsto (r + \kappa(\psi^{-1}(x)), \psi^{-1}(x))$ is the inverse of $\tilde{\psi}$, so $\hat{\psi}$ is a symplectomorphism.

Conversely, assume that $\tilde{\psi}$ is a symplectomorphism. By considering the projection of $\tilde{\psi}$ on M , we see that ψ is a diffeomorphism. Denote $\beta = e^{-\kappa}\psi^*\alpha$. For every $X \in \Gamma(TM)$,

$$d(e^r\alpha)(\partial_r, X) = e^rd\alpha(\partial_r, X) + e^rdr \wedge \alpha(\partial_r, X) = e^r\alpha(X),$$

hence $\tilde{\psi}^*d(e^r\alpha) = d(e^{r-\kappa(\psi)}\psi^*\alpha) = d(e^r\alpha)$ implies $e^r\beta(X) = e^r\alpha(X)$. As X is arbitrary, $\alpha = \beta$, i.e., $\psi^*\alpha = e^{\kappa}\alpha$. $\square$

Exercise 3.6. Given a contact manifold $(M, \xi = \ker \alpha)$, let h_t be the contact Hamiltonian function generating $\psi = \{\psi_t\}$ in $\text{Cont}_0(M, \xi)$ such that $\psi_t^*\alpha = e^{\kappa(\psi_t)}\alpha$, and g_t be the contact Hamiltonian function generating $\varphi = \{\varphi_t\}$ in $\text{Cont}_0(M, \xi)$. Then $h_t + (e^{\kappa(\psi_t)}g_t) \circ (\psi_t^{-1})$ generates the composition $\psi \circ \varphi = \{\psi_t \circ \varphi_t\}$. Also, the composition $\psi^{-1} \circ \varphi$ is generated by $(e^{-\kappa(\psi_t)}(g_t - h_t)) \circ \psi_t$.

Solution. Let X_t, Y_t be the vector fields generated by ψ, φ respectively. For every $x \in M$,

$$\frac{d}{dt}(\psi_t \circ \varphi_t)(x) = (\psi_t)_* \frac{d}{dt}\varphi_t(x) + \frac{d\psi_t}{dt}(\varphi_t(x)) = (\psi_t)_*(Y_t(\varphi_t(x))) + X_t(\psi_t \circ \varphi_t(x)).$$

Replacing x by $(\psi_t \circ \varphi_t)^{-1}(x)$, the vector field generated by $\psi \circ \varphi$ is $X_t + (\psi_t)_*Y_t$. It suffices to show that this vector field is generated by $(e^{\kappa(\psi_t)}g_t) \circ (\psi_t^{-1})$. Indeed,

$$f_t(x) = \alpha(X_t + (\psi_t)_*Y_t)(x) = h_t(x) + ((\psi_t^*\alpha)(Y_t))(\psi_t^{-1}(x)) = (h_t + (e^{\kappa(\psi_t)}g_t) \circ \psi_t^{-1})(x).$$

Direct computation yields for every $Z \in \xi_x$

$$\begin{aligned} d\alpha(X_t + (\psi_t)_*Y_t, Z)(x) &= -dh_t(Z)(x) + d(\psi_t^*\alpha)(Y_t, (\psi_t^{-1})_*Z)(\psi_t^{-1}(x)) \\ &= -dh_t(Z)(x) + \iota_{Y_t}d(e^{\kappa(\psi_t)}\alpha)((\psi_t^{-1})_*Z)(\psi_t^{-1}(x)). \end{aligned}$$

Since $\mathcal{L}_{Y_t}\alpha = dg_t(R_\alpha)\alpha$,

$$\mathcal{L}_{Y_t}\alpha((\psi_t^{-1})_*Z)(x) = dg_t(R_\alpha)(\psi_t^{-1})^*\alpha(Z)(\psi_t(x)) = 0.$$

Hence by Cartan's magic formula,

$$\begin{aligned} d\alpha(X_t + (\psi_t)_* Y_t, Z)(x) &= -dh_t(Z)(x) - d\iota_{Y_t}(e^{\kappa(\psi_t)}\alpha)((\psi_t^{-1})_* Z)(\psi_t^{-1}(x)) \\ &= -dh_t(Z)(x) - d(e^{\kappa(\psi_t)}g_t)((\psi_t^{-1})_* Z)(\psi_t^{-1}(x)) \\ &= -dh_t(Z)(x) - d(e^{\kappa(\psi_t)}g_t \circ \psi_t^{-1})(Z). \end{aligned}$$

The same equality holds for R_α , so this completes the verification.

As for the second conclusion, if $g_t = (e^{-\kappa(\psi_t)}h_t) \circ \psi_t$, then $h_t + (e^{\kappa(\psi_t)}g_t) \circ (\psi_t^{-1}) = 0$ generates the identity, so g_t generates $\{\psi_t^{-1}\}$. Then $\psi^{-1} \circ \varphi$ is generated by

$$(e^{-\kappa(\psi_t)}h_t) \circ \psi_t + (e^{-\kappa(\psi_t)}g_t) \circ \psi_t = (e^{-\kappa(\psi_t)}(g_t - h_t)) \circ \psi_t.$$

□

Exercise 3.7. For a contact manifold $(M, \xi = \ker \alpha)$, prove that the contact Poisson bracket $\{-, -\}_c$ is skew-symmetric and satisfies the Jacobi identity, i.e.

$$\begin{aligned} \{F, G\}_c + \{G, F\}_c &= 0, \\ \{F, \{G, H\}_c\}_c + \{G, \{H, F\}_c\}_c + \{H, \{F, G\}_c\}_c &= 0, \end{aligned}$$

for any $F, G, H \in C^\infty(M)$.

Solution. $C^\infty(M)$ is isomorphic to contact vector fields $\mathfrak{X}(M, \xi)$ as vector spaces via $H \mapsto X_H$. The inverse map is $X \mapsto \alpha(X)$. We now show that $\mathfrak{X}(M, \xi)$ is a Lie subalgebra of $\Gamma(TM)$. Indeed,

$$\begin{aligned} \mathcal{L}_{[X_F, X_G]}\alpha &= \mathcal{L}_{X_F}\mathcal{L}_{X_G}\alpha - \mathcal{L}_{X_G}\mathcal{L}_{X_F}\alpha \\ &= \mathcal{L}_{X_F}((R_\alpha G)\alpha) - \mathcal{L}_{X_G}((R_\alpha F)\alpha) \\ &= (X_F R_\alpha G)\alpha + (R_\alpha G)(R_\alpha F)\alpha - (X_G R_\alpha F)\alpha - (R_\alpha F)(R_\alpha G)\alpha \\ &= (X_F R_\alpha G - X_G R_\alpha F)\alpha. \end{aligned}$$

Then the flow generated by $[X_F, X_G]$ preserves ξ , so $[X_F, X_G]$ is a contact vector field. We see that

$$\{F, G\}_c = \alpha([X_F, X_G])$$

is the induced Lie bracket on $C^\infty(M)$. □

Exercise 3.8. Suppose $\varphi \in \text{Cont}(M, \xi = \ker \alpha)$ is a contactomorphism with conformal factor $\kappa(\varphi) : M \rightarrow \mathbb{R}$. Then for any $F, G \in C^\infty(M)$, we have the following equality,

$$\{e^{-\kappa(\varphi)}\varphi^*F, e^{-\kappa(\varphi)}\varphi^*G\}_c = e^{-\kappa(\varphi)}\varphi^*\{F, G\}_c.$$

Solution. Let X_F be the contact vector field generated by $F \in C^\infty(M)$. Then on SM ,

$$d(e^r\alpha)(X_F - dF(R_\alpha)\partial_r, \partial_r) = -e^r F.$$

For every $Y \in \Gamma(TM)$, we have

$$d(e^r\alpha)(X_F - dF(R_\alpha)\partial_r, Y) = -d(e^r F)(Y).$$

We see that $X_F - dF(R_\alpha)\partial_r$ is the Hamiltonian vector field on SM , generated by $e^r F$.

Denote $SX_f = X_f - df(R_\alpha)\partial_r$ for $f \in C^\infty(M)$. We have

$$\begin{aligned}\{e^r F, e^r G\} &= -d(e^r \alpha)(SX_F, SX_G) \\ &= e^r((R_\alpha F)G - (R_\alpha G)F - d\alpha(X_F, X_G)).\end{aligned}$$

As $d\alpha(X_F, X_G) = (\iota_{X_F} d\alpha)(X_G) = (R_\alpha F)G - X_G F$,

$$\{e^r F, e^r G\} = e^r(X_G F - (R_\alpha G)F) = e^r \{F, G\}_c.$$

Since $\tilde{\varphi}^*(e^r f) = e^{r-\kappa(\varphi)}\varphi^* f$ for $f \in C^\infty(M)$, we obtain

$$\begin{aligned}e^r \{e^{-\kappa(\varphi)}\varphi^* F, e^{-\kappa(\varphi)}\varphi^* G\}_c &= \{e^{r-\kappa(\varphi)}\varphi^* F, e^{r-\kappa(\varphi)}\varphi^* G\} \\ &= \{\tilde{\varphi}^*(e^r F), \tilde{\varphi}^*(e^r G)\} \\ &= \tilde{\varphi}^* \{e^r F, e^r G\} \\ &= \tilde{\varphi}^*(e^r \{F, G\}_c) \\ &= e^{-\kappa(\varphi)}\varphi^* \{F, G\}_c.\end{aligned}$$

□

Exercise 3.9. Given a $(2n+1)$ -dimensional contact manifold $(M, \xi = \ker \alpha)$, let L be a n -dimensional submanifold of M . Denote $I_L := \{F \in C^\infty(M) \mid F|_L = 0\}$. Prove that L is a Legendrian submanifold of (M, ξ) if and only if the contact Poisson bracket $\{-, -\}_c$ is closed in I_L , i.e. $\{F, G\}_c \in I_L$ for any $F, G \in I_L$.

Solution. Suppose L is Legendrian and $F, G \in I_L$. Then $\mathbb{R} \times L$ is Lagrangian in SM . Then by Exercise 3.8, $\{F, G\}_c = e^{-r} \{e^r F, e^r G\}$ on $\mathbb{R} \times L$. Since $e^r F, e^r G$ vanish on $\mathbb{R} \times L$, we have $\{F, G\}_c$ vanish on $\mathbb{R} \times L$, so $\{F, G\}_c \in I_L$.

Conversely, suppose I_L is closed under the contact Poisson bracket. By contradiction, suppose there exists $x \in L$ such that $(T_x L, d\alpha|_{\xi_x})$ is not coisotropic. We can find $v \in \xi_x$ such that $v \notin (T_x L)^\omega \setminus T_x L$. Then We can take $G \in I_L$ such that $d\alpha(X_G, v) = dG(R_\alpha)\alpha(v) - dG_x(v) = -dG(v) \neq 0$, which is always possible if we work in a local coordinate. In particular, $X_G \notin T_x L$. In the same way, we can construct $F \in I_L$ such that $d\alpha_x(X_F, X_G) = -dF(X_G)(x) \neq 0$. However,

$$\{F, G\}_c(x) = dF(X_G)(x) - dG(R_\alpha)F(x) = dF(X_G)(x) \neq 0,$$

which violates our hypothesis, so L must be Legendrian. □

Exercise 3.10. Given a contact manifold $(M, \xi = \ker \alpha)$ and $F \in C^\infty(M)$, let X_F be the contact Hamiltonian vector field of F , i.e.

$$\begin{aligned}F &= \alpha(X_F), \\ dF &= dF(R_\alpha)\alpha - \iota_{X_F} d\alpha.\end{aligned}$$

Prove that $[X_F, X_G] = X_{\{G, F\}_c}$ for any $F, G \in C^\infty(M)$.

Solution. The conclusion again follows from the Lie algebra isomorphism between $C^\infty(M)$ and contact vector fields $\mathfrak{X}(M, \xi)$. □

4 HOMEWORK 4

Exercise 4.1. Construct a smooth family of contact 1-forms $\{\alpha_t\}_{t \in [0,1]}$ on a smooth manifold M where there does not exist a smooth isotopy $\{\varphi_t\}_{t \in [0,1]} \subset \text{Diff}(M)$ such that $\varphi_t^* \alpha_t = \alpha_0$

Solution. Let $\alpha_t = (1+t)\alpha$ be a smooth family of contact forms on S^3 , where α is a given contact form, say that obtained as a convex hypersurface in $(\mathbb{R}^4, \omega_{\text{std}})$. If there exists a smooth isotopy $\{\varphi_t\}$ such that $\varphi_t^* \alpha_t = \alpha_0$, then

$$\int_{S^3} \alpha \wedge d\alpha = \int_{S^3} \alpha_0 \wedge d\alpha_0 = \int_{S^3} \varphi_t^* \alpha_t \wedge d\varphi_t^* \alpha_t = (1+t)^3 \int_{S^3} \alpha \wedge d\alpha.$$

We obtain $(1+t)^3 = 1$ for all $t \in [0,1]$, a contradiction. $\square$

Exercise 4.2. Prove that 2-dimensional closed symplectic manifolds denoted by (M_0, ω_0) and (M_1, ω_1) are symplectomorphic if and only if they have the same genus and area. Here, the area is computed by $\int_{M_i} \omega_i$ for $i = 0, 1$.

Solution. If (M_0, ω_0) and (M_1, ω_1) are symplectomorphic, then they are diffeomorphic (so they have the same genus) and $\int_{M_0} \omega_0 = \int_{M_1} \omega_1$.

Conversely, assume that (M_0, ω_0) and (M_1, ω_1) have the same genus and area. First, M_0 and M_1 are diffeomorphic (this can be done by applying uniformization theorem, for example). By choosing an orientation-preserving diffeomorphism, we may assume $M_0 = M_1 = M$. Then $\omega_t = (1-t)\omega_0 + t\omega_1$ is a smooth family of area form with constant area. Here ω_t is always non-degenerate because $\omega_i = f_i \eta$ for some fixed area form η and smooth positive functions f_i , and $\omega_t = ((1-t)f_0 + tf_1)\eta$. Since $H^2(M; \mathbb{R})$ is isomorphic to $\mathbb{R}$ via integration, by Moser's trick, we can find a smooth isotopy $\{\varphi_t\}$ such that $\varphi_t^* \omega_t = \omega_0$, and in particular, $\varphi_1^* \omega_1 = \omega_0$. $\square$

Exercise 4.3. Let D_1, D_2 be embedded closed disks in $(\mathbb{R}^2, \omega_{\text{std}})$ with the same area and $\gamma_i = \partial D_i$ for $i = 1, 2$. Assume that we can choose a orientation-preserving diffeomorphism $\phi \in \text{Diff}_c(\mathbb{R}^2)$ which sends D_1 to D_2 and ϕ preserves volume near γ_1 . Prove that γ_1 and γ_2 are Hamiltonian isotopic, i.e. there exists a Hamiltonian $H : [0, 1] \times \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $\varphi_H^1(\gamma_0) = \gamma_1$.

Solution. Let $\omega_0 = \omega_{\text{std}}, \omega_1 = \phi^* \omega_{\text{std}}, \omega_t = (1-t)\omega_0 + t\omega_1$. Now we find a compactly supported isotopy $\{\rho^t\}$ such that $\rho^t|_{\gamma_1} = \mathbb{1}$ and $(\rho^t)^* \omega_t = \omega_0$. We can find σ such that $d\sigma = \omega_1 - \omega_0$. Since ϕ preserves volume near γ_1 , $\omega_1 = \omega_0$ near γ_1 outside a compact set and we can require σ to vanish near γ_1 and outside a compact set. Then we can pick X_t making $\iota_{X_t} \omega_t + \sigma = 0$. By Moser's trick, we can let $\{\rho^t\}$ be the flow generated by X_t . Then $\psi \circ \rho^1$ is a compactly supported symplectomorphism sending D_1 to D_2 .

Let f be a compactly supported symplectomorphism of $(\mathbb{R}^2, \omega_{\text{std}})$. By adding suitable compactly supported translation and rotation, we may assume f fixes the origin and $df_0 = \mathbb{1}$. Then

$$f_t(x) = \frac{1}{t} f(tx)$$

is also a compactly supported symplectomorphism for $0 < t \leq 1$ with $f_1 = f$. Moreover, $f_t \rightarrow df_0 = 1$ as $t \rightarrow 0^+$. In particular, $\text{Symp}_c(\mathbb{R}^2, \omega_{\text{std}})$ is path connected. Due to the theorem by Banyaga and Oh, $\text{Symp}_c(\mathbb{R}^2, \omega_{\text{std}}) = \text{Ham}_c(\mathbb{R}^2, \omega_{\text{std}})$. $\square$

Exercise 4.4. Let $(M, \xi = \ker \alpha)$ be a contact manifold and $L \subset M$ be a Legendrian submanifold. Prove that there exists a neighborhood U of L in M , a neighborhood V of $0_L \times \{0\}$ in $T^*L \times \mathbb{R}$, and a diffeomorphism $\varphi : V \rightarrow U$ such that $\varphi^* \alpha = \lambda_{\text{can}} + dt$, where t is the $\mathbb{R}$ -coordinate. Here, $T^*L \times \mathbb{R}$ is a contact manifold with contact form $\lambda_{\text{can}} + dt$ where λ_{can} is the canonical 1-form locally defined by $\sum_{i=1}^n p_i dq_i$.

Solution. Say $\dim M = 2n + 1$. Let R_α be the Reeb vector field of α . Then R_α induces the splitting $TM = E \oplus \mathbb{R}$, where E is a bundle of rank $2n$ determined by ξ and $\mathbb{R} \rightarrow M$ is a trivial line bundle. Choose a metric g on E and split $E|_L = TL \oplus F$. Here $F = TL^\perp$. By declaring R_α to be a unit vector orthogonal to E , we extend g to a Riemannian metric on TM . Then under g , we see that the normal bundle of L in M is isomorphic to $T^*L \oplus \mathbb{R}$. Note that $T^*L \oplus \mathbb{R} \simeq T^*L \times \mathbb{R}$, $(x, v, t) \mapsto ((x, v), t)$.

By tubular neighborhood theorem, we can find a neighborhood U of L in M , a neighborhood V of $0_L \times \{0\}$ in $T^*L \times \mathbb{R}$ and a diffeomorphism $\psi : U \rightarrow V$. Let $\beta = \psi^*(\lambda_{\text{can}} + dt)$ and consider $\alpha_s = (1 - s)\alpha + s\beta$. Note that by our construction, $d\psi_x$ is an isomorphism $T_x L \oplus F_x \oplus \mathbb{R} \simeq T_x L \oplus T_x^* L \oplus \mathbb{R}$ respecting the direct sum decomposition, so

$$\begin{aligned} \alpha(x) &= \beta(x), \\ \alpha_s \wedge (d\alpha_s)^n(x) &= \alpha \wedge (d\alpha)^n(x). \end{aligned}$$

By shrinking U, V if necessary, α_s is a contact 1-form on U . Now we need to find an isotopy $\{\psi^s\}$ preserving L , with $(\psi^s)^* \alpha_s = \alpha$. Let X_s be the vector field generated by $\{\psi^s\}$ and then X_s satisfies the equation

$$0 = \frac{d}{ds}(\psi^s)^* \alpha_s = (\psi^s)^*(\beta - \alpha + \mathcal{L}_{X_s} \alpha_s),$$

i.e. $\alpha - \beta = \mathcal{L}_{X_s} \alpha_s$. We now solve this equation. Let X_s lies in $\ker \alpha_s$. Then $\mathcal{L}_{X_s} \alpha_s = \iota_{X_s} d\alpha_s$. Since $d\alpha_s$ is non-degenerate on $\ker \alpha_s$, such X_s exists. Moreover, as $\alpha - \beta$ vanishes on L , so does X_s . Finally, we take $\varphi = (\psi \circ \psi^1)^{-1}$. Then

$$(\varphi^{-1})^*(\lambda_{\text{can}} + dt) = (\psi^1)^* \psi^*(\lambda_{\text{can}} + dt) = (\psi^1)^* \beta = (\psi^1)^* \alpha_1 = \alpha.$$

$\square$

5 MIDTERM EXAM

Problem A. [Moser's trick]

- (a) Give a counterexample in dimension 4: Find two symplectic forms on $\mathbb{R}^4$ that induce the same orientation while a convex combination of them is non-degenerate.
- (b) Prove the following version of Morse lemma via Moser's trick: Let φ_1 and φ_2 be two smooth functions on $\mathbb{R}^n$ such that $\varphi_1(0) = \varphi_2(0) = 0$, $\nabla\varphi_1(0) = \nabla\varphi_2(0) = 0$, and the Hessian matrix $\left(\frac{\partial^2\varphi_i}{\partial x_i\partial x_j}\right)(0) = \left(\frac{\partial^2\varphi_2}{\partial x_i\partial x_j}\right)(0)$ is non-degenerate. Prove: There exist neighborhoods U_1 and U_2 of $\mathbb{R}^n$ and a diffeomorphism $f: U_1 \rightarrow U_2$ such that $f(0) = 0$ and $f^*(\varphi_2|_{U_2}) = \varphi_1|_{U_1}$.

Solution. (a) Let x_1, y_1, x_2, y_2 be coordinates of $\mathbb{R}^4$. Consider

$$\begin{aligned}\omega_0 &= dx_1 \wedge dy_1 + dx_2 \wedge dy_2 + dx_1 \wedge dx_2, \\ \omega_1 &= dx_1 \wedge dy_1 + dx_2 \wedge dy_2 + 4dy_1 \wedge dy_2.\end{aligned}$$

Then $\omega_0^2 = \omega_1^2 = 2dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$, so they are symplectic forms inducing the same orientation. Let

$$\omega_t = (1-t)\omega_0 + t\omega_1 = dx_1 \wedge dy_1 + dx_2 \wedge dy_2 + (1-t)dx_1 \wedge dx_2 + 4tdy_1 \wedge dy_2.$$

We have $\omega_t^2 = 2(2t-1)^2 dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$, so ω_t is degenerate if $t = \frac{1}{2}$.

(b) Let $\varphi_t = (1-t)\varphi_1 + t\varphi_2$. Our strategy is to find an isotopy $\{f_t\}_{t \in [0,1]}$ such that $f_t(0) = 0$ and $f_t^*\varphi_t = \varphi_2$ on some open neighborhood of 0. Equation $f_t^*\varphi_t = \varphi_2$ is equivalent to

$$0 = \frac{d}{dt}f_t^*\varphi_t = f_t^*(\dot{\varphi}_t + \mathcal{L}_{X_t}\varphi_t),$$

where X_t is the vector field generated by $\{f_t\}$. In particular, it suffices to solve the equation $\mathcal{L}_{X_t}\varphi_t = -\dot{\varphi}_t = \varphi_1 - \varphi_2$ for X_t . Take $X_t = \sum A_i(x, t)\partial_{x_i}$. Then our problem becomes: find functions $A_i(x, t)$ with $A_i(0, t) = 0$ such that

$$\sum_i A_i(x, t) \frac{\partial \varphi_t}{\partial x_i} = \varphi_1 - \varphi_2.$$

Since $\left(\frac{\partial^2\varphi_t}{\partial x_i\partial x_j}\right)(0) = \left(\frac{\partial^2\varphi_2}{\partial x_i\partial x_j}\right)(0)$ is non-degenerate, $u(x, t) = \frac{\partial \varphi_t}{\partial x_i}$ is a coordinate change near 0 with $u(0, t) = 0$. Under coordinates $u_1, \dots, u_n$, the equation becomes

$$\varphi_1(x(u, t)) - \varphi_2(x(u, t)) = \sum_i A_i(x(u, t), t)u_i.$$

Let $H(u, t) = \varphi_1(x(u, t)) - \varphi_2(x(u, t))$. Then

$$H = \int_0^1 \frac{d}{ds} H(su, t) ds = \sum_i u_i g_i(u, t),$$

where

$$g_i(u, t) = \int_0^1 \frac{\partial H}{\partial u_i}(su, t) ds.$$

As $g_i(0, t) = \frac{\partial H}{\partial u_i}(0, t) = 0$, we can take $A_i(x, t) = g_i(u(x, t), t)$. This proves the existence of $\{f_t\}$. $\square$

Problem B. [Calabi invariant] Let $\mathbb{D} = \{z \in \mathbb{C} \mid |z| \leq 1\}$ and $\omega = \frac{1}{2\pi} r dr \wedge d\theta$ for polar coordinates (r, θ) . Let $C_c^\infty(S^1 \times \mathbb{D})$ be the set of time-dependent smooth functions of $\mathbb{D}$ whose supports are contained in the interior of $\mathbb{D}$. Let $\text{Ham}_c(\mathbb{D}, \omega) = \{\varphi_H^1 \mid H \in C_c^\infty(S^1 \times \mathbb{D})\}$, the group of compactly supported Hamiltonian diffeomorphisms of $(\mathbb{D}, \omega)$.

(a) Assume that Calabi invariant

$$\text{Cal}: \text{Ham}_c(\mathbb{D}, \omega) \rightarrow \mathbb{R}, \varphi_H^1 \mapsto \int_{S^1} dt \int_{\mathbb{D}} H \omega,$$

is well-defined. Prove that Cal is a group homomorphism.

(b) Let $f: [0, 1] \rightarrow \mathbb{R}$ be a smooth function which vanishes near 1. Define a diffeomorphism ϕ_f of $\mathbb{D}$ by $\varphi_f(0) = 0$ and $\varphi_f(r, \theta) = (r, \theta + 2\pi f(r))$. Prove that $\phi_f \in \text{Ham}_c(\mathbb{D}, \omega)$ and

$$\text{Cal}(\phi_f) = - \int_0^1 r dr \int_r^1 s f(s) ds.$$

Let $\text{Homeo}_c(\mathbb{D}, \omega)$ be the group of homeomorphisms of $\mathbb{D}$ which preserve the measure induced by ω and are identity near the boundary. For any $H \in C^\infty(S^1 \times \mathbb{D})$, the Hofer norm of H is defined by

$$\|H\|_{\text{Hofer}} = \int_0^1 \left(\max_{x \in \mathbb{D}} H(t, \cdot) - \min_{x \in \mathbb{D}} H(t, \cdot) \right) dt.$$

An element $\varphi \in \text{Homeo}_c(\mathbb{D}, \omega)$ is called a finite-energy homeomorphism if there exists a sequence $\{H_i\}$ in $C_c^\infty(S^1 \times \mathbb{D})$ such that the sequence $\{\|H_i\|_{\text{Hofer}}\}$ is bounded above and the Hamiltonian diffeomorphisms $\varphi_{H_i}^1$ converge uniformly to φ as $i \rightarrow \infty$. Let $\text{FHomeo}_c(\mathbb{D}, \omega)$ be the set of all finite-energy homeomorphisms.

(c) Prove that $\text{FHomeo}_c(\mathbb{D}, \omega)$ is a normal subgroup of $\text{Homeo}_c(\mathbb{D}, \omega)$. Hint: You may use the fact that for any $\psi \in \text{Homeo}_c(\mathbb{D}, \omega)$, there exists a sequence $\{\psi_i\}$ in $\text{Ham}_c(\mathbb{D}, \omega)$ which converges uniformly to ψ .

Solution. (a) Let $\varphi, \psi \in \text{Ham}_c(\mathbb{D}, \omega)$. By definition, we can find $H_t, G_t \in C_c^\infty(S^1 \times \mathbb{D})$ generating Hamiltonian flows $\{\varphi^t\}, \{\psi^t\}$ such that $\varphi^1 = \varphi, \psi^1 = \psi$. The product $\{\varphi^t \psi^t\}$ is generated by $H_t + G_t \circ (\varphi^t)^{-1}$. Note that $H_t + G_t \circ (\varphi^t)^{-1} \in C_c^\infty(S^1 \times \mathbb{D})$,

hence

$$\begin{aligned}
\text{Cal}(\varphi\psi) &= \int_{S^1} dt \int_{\mathbb{D}} H_t + G_t \circ (\varphi^t)^{-1} \omega \\
&= \int_{S^1} dt \int_{\mathbb{D}} H_t \omega + \int_{S^1} dt \int_{\mathbb{D}} G_t \circ (\varphi^t)^{-1} \omega \\
&= \int_{S^1} dt \int_{\mathbb{D}} H_t \omega + \int_{S^1} dt \int_{\mathbb{D}} G_t (\varphi^t)^* \omega \\
&= \int_{S^1} dt \int_{\mathbb{D}} H_t \omega + \int_{S^1} dt \int_{\mathbb{D}} G_t \omega \\
&= \text{Cal}(\varphi) + \text{Cal}(\psi).
\end{aligned}$$

(b) Consider

$$H(r, \theta) = - \int_r^1 s f(s) ds.$$

We have $H \in C_c^\infty(S^1 \times \mathbb{D})$. Let $A\partial_r + B\partial_\theta$ be the Hamiltonian vector field generated by H . As $dH = rf(r)dr$,

$$\iota_X \omega = \frac{1}{2\pi} r (Ad\theta - Bdr) = -rf(r)dr.$$

We obtain $X = 2\pi f(r)\partial_\theta$, hence the Hamiltonian flow generated by H is

$$\varphi_H^t(r, \theta) = (r, \theta + 2\pi f(r)t).$$

In particular, $\phi_f = \varphi_H^1 \in \text{Ham}_c(\mathbb{D}, \omega)$ and

$$\begin{aligned}
\text{Cal}(\phi_f) &= \int_{S^1} dt \int_{\mathbb{D}} H \omega \\
&= \int_{\mathbb{D}} H \omega \\
&= \int_{[0,1] \times [0,2\pi]} H(r) \frac{1}{2\pi} r dr \wedge d\theta \\
&= \int_0^1 H(r) r dr \int_0^{2\pi} \frac{1}{2\pi} d\theta \\
&= - \int_0^1 r dr \int_r^1 s f(s) ds.
\end{aligned}$$

(c) As a priori, we need to show that $\text{FHomeo}_c(\mathbb{D}, \omega)$ is a subgroup of $\text{Homeo}_c(\mathbb{D}, \omega)$. As $\mathbb{1}$ is generated by $0 \in C_c^\infty(S^1 \times M)$ and $\|0\|_{\text{Hofer}} = 0$, $\mathbb{1} \in \text{FHomeo}_c(\mathbb{D}, \omega)$. Let $f, g \in \text{FHomeo}_c(\mathbb{D}, \omega)$. There exists two sequences $\{F_i\}, \{G_i\}$ in $C_c^\infty(S^1 \times \mathbb{D})$ such that $\{\|F_i\|_{\text{Hofer}}\}, \{\|G_i\|_{\text{Hofer}}\}$ are both bounded above and $\{\varphi_{F_i}^1\}, \{\varphi_{G_i}^1\}$ converge uniformly to f, g respectively. Then $\{\varphi_{F_i}^t \varphi_{G_i}^t\}$ is generated by $F_i + G_i \circ (\varphi_{F_i}^t)^{-1} \in C_c^\infty(S^1 \times M)$. Moreover, from the definition of Hofer norm we see that $\|F + G\|_{\text{Hofer}} \leq \|F\|_{\text{Hofer}} +$

$\|G\|_{\text{Hofer}}$ and $\|F \circ h\|_{\text{Hofer}} = \|F\|_{\text{Hofer}}$ for all $h \in \text{Diff}(M)$, so $\|F_i + G_i \circ (\varphi_{F_i}^t)^{-1}\|_{\text{Hofer}}$ is bounded above:

$$\|F_i + G_i \circ (\varphi_{F_i}^t)^{-1}\|_{\text{Hofer}} \leq \|F_i\|_{\text{Hofer}} + \|G_i \circ (\varphi_{F_i}^t)^{-1}\|_{\text{Hofer}} = \|F_i\|_{\text{Hofer}} + \|G_i\|_{\text{Hofer}}.$$

As $\{\varphi_{F_i}^1 \varphi_{G_i}^1\}$ converges uniformly to fg , $fg \in \text{FHomeo}_c(\mathbb{D}, \omega)$. We also need to show that $\text{FHomeo}_c(\mathbb{D}, \omega)$ is closed under inverse. Indeed, $\{(\varphi_{F_i}^t)^{-1}\}$ is generated by $-F_i \circ f \in C_c^\infty(S^1 \times M)$. $\|-F_i \circ f\|_{\text{Hofer}} = \|F_i\|_{\text{Hofer}}$ and $\{(\varphi_{F_i}^1)^{-1}\}$ converges to f^{-1} uniformly, so $f^{-1} \in \text{FHomeo}_c(\mathbb{D}, \omega)$.

It remains to show that $\text{FHomeo}_c(\mathbb{D}, \omega)$ is normal in $\text{Homeo}_c(\mathbb{D}, \omega)$. Let $\varphi \in \text{FHomeo}_c(\mathbb{D}, \omega)$ and $\psi \in \text{Homeo}_c(\mathbb{D}, \omega)$. We can find a sequence $\{\psi_i\}$ in $\text{Ham}_c(\mathbb{D}, \omega)$ which converges uniformly to ψ . By definition, there exists a sequence $\{H_i\}$ in $C_c^\infty(S^1 \times \mathbb{D})$ such that $\{\|H_i\|_{\text{Hofer}}\}$ is bounded above and $\{\varphi_{H_i}^1\}$ converges uniformly to φ . Then $\psi_i^{-1} \varphi_{H_i}^1 \psi_i$ is a compactly supported Hamiltonian diffeomorphism generated by $H_i \circ \psi_i \in C_c^\infty(S^1 \times M)$, with $\{\psi_i^{-1} \varphi_{H_i}^1 \psi_i\}$ uniformly converging to $\psi^{-1} \varphi \psi$. $\|H_i \circ \psi_i\|_{\text{Hofer}} = \|H_i\|_{\text{Hofer}}$ is bounded above, so $\psi^{-1} \varphi \psi \in \text{FHomeo}_c(\mathbb{D}, \omega)$. $\square$

Problem C. [Billiard system] Let D be a convex embedded disc in $\mathbb{R}^2$ with a smooth boundary $\partial D = \gamma$. We parametrize the curve counterclockwise by arc length, i.e. $\gamma: \mathbb{R}/L\mathbb{Z} \rightarrow \mathbb{R}^2$ for some positive L such that $|\dot{\gamma}(t)| = 1$ for all $t \in \mathbb{R}/L\mathbb{Z}$.

(a) For $0 \leq t_1 < t < t_2 < L$, we define

$$L_{t_1, t_2}(t) = |\gamma(t) - \gamma(t_1)| + |\gamma(t_2) - \gamma(t)|.$$

- Prove that t is a critical point of L_{t_1, t_2} if and only if the incidence angle θ_1 between $\gamma(t) - \gamma(t_1)$ and $\dot{\gamma}(t)$ is equal to the reflection angle θ_2 between $\dot{\gamma}(t)$ and $\gamma(t_2) - \gamma(t)$.
- (b) Let $M = \{(t, \alpha) \mid t \in \mathbb{R}/L\mathbb{Z}, \alpha \in (0, \pi)\}$ be the phase space of billiard system of D which can be explained as follows. For each (t, α) , we consider $(\gamma(t), v(t)) \in \mathbb{R}^2 \times \mathbb{R}^2$ where the vector $v(t)$ is obtained by rotating $\dot{\gamma}(t)$ counterclockwise by angle α . Then, we have (t', α') where $\gamma(t')$ is the intersection point of γ and $\{\gamma + sv(t)\}_{s \in \mathbb{R}}$, and α' is the incidence angle between $v(t)$ and $\dot{\gamma}(t')$. Define the billiard map

$$T: M \rightarrow M, (t, \alpha) \mapsto (t', \alpha').$$

Prove that T preserves the symplectic form $\sin(\alpha)d\alpha \wedge dt$.

Solution. (a) Write $\gamma(t) = (x(t), y(t))$. Then

$$L_{t_1, t_2}(t) = \sqrt{(x(t) - x(t_1))^2 + (y(t) - y(t_1))^2} + \sqrt{(x(t_2) - x(t))^2 + (y(t_2) - y(t))^2}.$$

We have

$$\begin{aligned} \frac{d}{dt} L_{t_1, t_2}(t) &= \frac{\dot{x}(t)((x(t) - x(t_1)) + \dot{y}(t)(y(t) - y(t_1)))}{|\gamma(t) - \gamma(t_1)|} \\ &\quad + \frac{\dot{x}(t)((x(t) - x(t_2)) + \dot{y}(t)(y(t) - y(t_2)))}{|\gamma(t) - \gamma(t_2)|} \\ &= \frac{\dot{\gamma}(t) \cdot (\gamma(t) - \gamma(t_1))}{|\gamma(t) - \gamma(t_1)|} + \frac{\dot{\gamma}(t) \cdot (\gamma(t) - \gamma(t_2))}{|\gamma(t) - \gamma(t_2)|} \\ &= \cos \theta_1 - \cos \theta_2. \end{aligned}$$

As $\theta_1, \theta_2 \in (0, \frac{\pi}{2})$, t is a critical point of L_{t_1, t_2} if and only if $\theta_1 = \theta_2$.

(b) As priori, $\omega = \sin(\alpha)d\alpha \wedge dt$ is an area form as $\sin(\alpha) > 0$, so it is also symplectic. Consider the function $f = |\gamma(t') - \gamma(t)|$. We have

$$\begin{aligned}\frac{\partial}{\partial t}f &= -\cos \alpha, \\ \frac{\partial}{\partial t'}f &= \cos \alpha'.\end{aligned}$$

It follows that

$$df = \cos \alpha' dt' - \cos \alpha dt.$$

Note that ω admits a primitive 1-form $\lambda = -\cos(\alpha)dt$. Then $df = \lambda - T^*\lambda$, hence

$$0 = d(df) = d\lambda - dT^*\lambda = d\lambda - T^*d\lambda = \omega - T^*\omega.$$

□

Problem D. [Contact manifold and its symplectization] Let $(M^{2n+1}, \xi = \ker \alpha)$ be a contact manifold.

- (a) Let ψ be a contactomorphism of $(M, \xi = \ker \alpha)$ with $\psi^*\alpha = e^{\kappa(\psi)}\alpha$ where $\kappa(\psi): M \rightarrow \mathbb{R}$. If M is compact, prove that the set $(\kappa(\psi))^{-1}(0)$ is not empty. Hint: Consider $\int_M \alpha \wedge (d\alpha)^n$.

Consider the symplectization $(\mathbb{R} \times M, d(e^r \alpha))$ and a smooth function $h: \mathbb{R} \rightarrow \mathbb{R}$. Define an autonomous Hamiltonian H on $\mathbb{R} \times M$ by

$$H: \mathbb{R} \times M, (r, x) \mapsto h(e^r).$$

- (b) Prove that $X_H(r, x) = h'(e^r)R_\alpha(x)$ where R_α is the Reeb vector field of α .
(c) Let J be an almost complex structure on $\mathbb{R} \times M$ such that J preserves $\xi = \ker \alpha$ and $J(\partial_r) = R_\alpha$. Assume that $u(s, t) = (r(s, t), x(s, t)): \mathbb{R} \times S^1 \rightarrow \mathbb{R} \times M$

$$\frac{\partial u}{\partial s} + J(u) \left(\frac{\partial u}{\partial t} - X_H(u) \right) = 0.$$

Then, prove the following equation:

$$\Delta(e^r) = (d(e^r \alpha)) \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right) - h''(e^r) e^{2r} \frac{\partial r}{\partial s},$$

where $\Delta = (\partial_s)^2 + (\partial_t)^2$.

Solution. (a)

$$\int_M \alpha \wedge (d\alpha)^n = \int_M \psi^*(\alpha \wedge (d\alpha)^n) = \int_M e^{(n+1)\kappa(\psi)} \alpha \wedge (d\alpha)^n.$$

Since $\alpha \wedge (d\alpha)^n$ is a volume form, we have $e^{(n+1)\kappa(\psi)(x)} = 1$ for some $x \in M$, i.e. $\kappa(\psi)(x) = 0$.

(b) $d(e^r \alpha) = e^r(dr \wedge \alpha + d\alpha)$. For $X \in \xi_x$,

$$\begin{aligned} d(e^r \alpha)(R_\alpha, X) &= e^r(dr(R_\alpha)\alpha(X) - dr(X)\alpha(R_\alpha) - d\alpha(R_\alpha, X)) = 0, \\ d(e^r \alpha)(R_\alpha, R_\alpha) &= e^r(dr(R_\alpha)\alpha(R_\alpha) - dr(R_\alpha)\alpha(R_\alpha) - d\alpha(R_\alpha, R_\alpha)) = 0. \end{aligned}$$

We see that $d(e^r \alpha)(h'(e^r)R_\alpha, Y) = 0$ for all $Y \in \Gamma(TM)$. As ∂_r vanishes for $\Omega^*(M)$,

$$d(e^r \alpha)((h'(e^r)R_\alpha, \partial_r) = -e^r dr(\partial_r)\alpha(h'(e^r)R_\alpha) = -e^r h'(e^r) = -\partial_r H.$$

It follows that $d(e^r \alpha)(h'(e^r)R_\alpha, \cdot) = -dH$, so the conclusion follows.

(c) Let $\pi: X \mapsto X - \alpha(X)R_\alpha$ be the projection onto ξ . Since

$$\begin{aligned} \frac{\partial u}{\partial s} &= \frac{\partial r}{\partial s} \partial_r + \alpha\left(\frac{\partial x}{\partial s}\right) R_\alpha + \pi\left(\frac{\partial x}{\partial s}\right), \\ \frac{\partial u}{\partial t} &= \frac{\partial r}{\partial t} \partial_r + \alpha\left(\frac{\partial x}{\partial t}\right) R_\alpha + \pi\left(\frac{\partial x}{\partial t}\right). \end{aligned}$$

so we have

$$\begin{aligned} 0 &= \frac{\partial u}{\partial s} + J(u) \left(\frac{\partial u}{\partial t} - X_H(u) \right) \\ &= \left(\frac{\partial r}{\partial s} - \alpha\left(\frac{\partial x}{\partial t}\right) + h'(e^r) \right) \partial_r + \left(\alpha\left(\frac{\partial x}{\partial s}\right) + \frac{\partial r}{\partial t} \right) R_\alpha + \left(\pi\frac{\partial x}{\partial s} + J\pi\frac{\partial x}{\partial t} \right). \end{aligned}$$

We obtain

$$\begin{cases} \frac{\partial r}{\partial s} - \alpha\left(\frac{\partial x}{\partial t}\right) + h'(e^r) = 0, \\ \alpha\left(\frac{\partial x}{\partial s}\right) + \frac{\partial r}{\partial t} = 0, \\ \pi\frac{\partial x}{\partial s} + J\pi\frac{\partial x}{\partial t} = 0. \end{cases}$$

Direct computation yields

$$\Delta(e^r) = e^r \left(\Delta r + \left(\frac{\partial r}{\partial s} \right)^2 + \left(\frac{\partial r}{\partial t} \right)^2 \right).$$

We have

$$\begin{aligned} \Delta r &= \frac{\partial}{\partial s} \left(\frac{\partial r}{\partial s} \right) + \frac{\partial}{\partial t} \left(\frac{\partial r}{\partial t} \right) \\ &= \frac{\partial}{\partial s} \left(\alpha\left(\frac{\partial x}{\partial t}\right) - h'(e^r) \right) - \frac{\partial}{\partial t} \left(\alpha\left(\frac{\partial x}{\partial s}\right) \right) \\ &= \frac{\partial}{\partial s} x^* \alpha\left(\frac{\partial}{\partial t}\right) - \frac{\partial}{\partial t} x^* \alpha\left(\frac{\partial}{\partial s}\right) - h''(e^r) e^r \frac{\partial r}{\partial s} \\ &= d(x^* \alpha)\left(\frac{\partial}{\partial s}, \frac{\partial}{\partial t}\right) - h''(e^r) e^r \frac{\partial r}{\partial s} \\ &= d\alpha\left(\frac{\partial x}{\partial s}, \frac{\partial x}{\partial t}\right) - h''(e^r) e^r \frac{\partial r}{\partial s}, \end{aligned}$$

where the fourth equality holds because $[\partial_s, \partial_t] = 0$.

On the other hand, $(d(e^r \alpha)) \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right) = e^r (dr \wedge \alpha + d\alpha) \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right)$. For the first term in the bracket,

$$\begin{aligned} dr \wedge \alpha \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right) &= dr \left(\frac{\partial u}{\partial s} \right) \alpha \left(J \frac{\partial u}{\partial s} \right) - \alpha \left(\frac{\partial u}{\partial s} \right) dr \left(J \frac{\partial u}{\partial s} \right) \\ &= \left(\frac{\partial r}{\partial s} \right)^2 + \left(\frac{\partial r}{\partial t} \right)^2. \end{aligned}$$

For the second term in the bracket,

$$d\alpha \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right) = d\alpha \left(\frac{\partial x}{\partial s}, J \pi \frac{\partial x}{\partial s} \right) = d\alpha \left(\frac{\partial x}{\partial s}, \pi \frac{\partial x}{\partial t} \right) = d\alpha \left(\frac{\partial x}{\partial s}, \frac{\partial x}{\partial t} \right).$$

We see that

$$\begin{aligned} \Delta(e^r) &= e^r \left(\left(\frac{\partial r}{\partial s} \right)^2 + \left(\frac{\partial r}{\partial t} \right)^2 + d\alpha \left(\frac{\partial x}{\partial s}, \frac{\partial x}{\partial t} \right) - h''(e^r) e^r \frac{\partial r}{\partial s} \right) \\ &= (d(e^r \alpha)) \left(\frac{\partial u}{\partial s}, J \frac{\partial u}{\partial s} \right) - h''(e^r) e^{2r} \frac{\partial r}{\partial s}. \end{aligned}$$

□

Bonus Problem (Continued from Problem B). [Simplicity conjecture] Let $f: (0, 1] \rightarrow \mathbb{R}$ be a smooth function which vanishes near 1 and satisfies $\lim_{r \rightarrow 0} f(r) = \infty$. Let $\phi_f \in \text{Homeo}_c(\mathbb{D}, \omega)$ by $\phi_f(0) = 0$ and $\phi(r, \theta) = (r, \theta + 2\pi f(r))$. Prove that if

$$\int_0^1 r dr \int_r^1 s f(s) ds = +\infty,$$

then $\phi_f \notin \text{FHomeo}_c(\mathbb{D}, \omega)$. (This directly implies that the group $\text{Homeo}_c(\mathbb{D}, \omega)$ is not simple.)

For the proof of the bonus problem, you can use the following FACT:

(Michael Hutchings, Daniel Cristofaro-Gardiner, Vincent Humilière and Sobhan Seyfaddini) There are C^0 -continuous spectral invariants $c_d: \text{Homeo}_c(\mathbb{D}, \omega) \rightarrow \mathbb{R}$ satisfying the following properties for $d \in \mathbb{Z}_{\geq 1}$:

- (1) (Monotonicity) For $H, G \in C_c(S^1 \times \mathbb{D})$ with $H \leq G$, $c_d(\varphi_H^1) \leq c_d(\varphi_G^1)$.
- (2) (Hofer Continuity) For $H, G \in C_c^\infty(S^1 \times \mathbb{D})$, $|c_d(\varphi_H^1) - c_d(\varphi_G^1)| \leq d \|H - G\|_{\text{Hofer}}$.
- (3) (Calabi property) For a smooth function $f: [0, 1] \rightarrow \mathbb{R}$ which vanishes near 1 and near $\phi_f \in \text{Ham}_c(\mathbb{D}, \omega)$,

$$\lim_{d \rightarrow \infty} \frac{c_d(\phi_f)}{d} = -\text{Cal}(\phi_f).$$

Solution. As $f(r) \rightarrow \infty$ as $r \rightarrow 0$, we can pick smooth functions $f_i: [0, 1] \rightarrow \mathbb{R}$ such that $f_i = f$ on $[a_i, 1]$ and $f_i \leq f_{i+1}$, where $0 < a_{i+1} \leq a_i$ and $a_i \rightarrow 0$. Then

$$\text{Cal}(\phi_{f_i}) = - \int_0^1 r dr \int_r^1 s f_i(s) ds \leq - \int_{a_i}^1 r dr \int_r^1 s f_i(s) ds = - \int_{a_i}^1 r dr \int_r^1 s f(s) ds.$$

It follows that $\text{Cal}(\phi_{f_i}) \rightarrow -\infty$ as $i \rightarrow \infty$. By the Monotonicity property, we have $c_d(\phi_{f_i}) \leq c_d(\phi_{f_{i+1}})$. Since $\{\phi_{f_i}\}$ converges to ϕ_f in C^0 -topology, $c_d(\phi_f) = \lim_{i \rightarrow \infty} c_d(\phi_{f_i})$ and $c_d(\phi_f) \geq c_d(\phi_{f_i})$.

By contradiction, suppose $\phi_f \in \text{FHomeo}_c(\mathbb{D}, \omega)$. There exists a sequence $\{H_i\}$ in $C_c^\infty(S^1 \times \mathbb{D})$ such that $\{\|H_i\|_{\text{Hofer}}\}$ is bounded above by $C > 0$ and $\{\varphi_{H_i}^1\}$ converges uniformly to ϕ_f . Then

$$|c_d(\varphi_{H_i}^1) - c_d(\mathbb{1})| \leq d\|H_i\|_{\text{Hofer}} \leq dC,$$

so $c_d(\varphi_{H_i}^1) \leq |c_d(\mathbb{1})| + dC$. We see that

$$\frac{c_d(\phi_{f_i})}{d} \leq \frac{c_d(\phi_f)}{d} \leq \frac{|c_d(\mathbb{1})|}{d} + C,$$

so taking $d \rightarrow \infty$ we obtain

$$-\text{Cal}(\phi_{f_i}) = \lim_{d \rightarrow \infty} \frac{c_d(\phi_{f_i})}{d} \leq C.$$

This violates $\text{Cal}(\phi_{f_i}) \rightarrow -\infty$. □

6 HOMEWORK5

Exercise 6.1. Given a Morse function F on a smooth manifold X with a metric g , prove that for any critical point $x \in \text{Crit}(F)$, the moduli space $\widetilde{\mathcal{M}}_{F,g}(x, x)$ contains only the constant flow at point x .

Solution. Let $u(t)$ be a flowline with $u(-\infty) = u(\infty) = x$. Since

$$\frac{d}{dt}F(u(t)) = dF_{u(t)}(\dot{u}(t)) = g_{u(t)}(\nabla F, \dot{u}(t)) = -g_{u(t)}(\nabla F, \nabla F) \leq 0,$$

$F(u(-\infty)) = F(u(\infty)) = F(x)$ implies $\frac{d}{dt}F(u) = 0$, i.e. $\nabla F(u(t)) = 0$ for all t . It follows that $u(t) \in \text{Crit}(F)$ for all t . As F is Morse, $\text{Crit}(F)$ is discrete and we obtain $u(t) = x$. $\square$

Exercise 6.2. Consider the following function $f : \mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{R}$ defined by

$$f(z_0, \dots, z_n) = \frac{\sum_{j=0}^n j |z_j|^2}{\sum_{j=0}^n |z_j|^2}, \quad (z_0, \dots, z_n) \in \mathbb{C}^{n+1} \setminus \{0\}.$$

Recall that $\mathbb{CP}^n$ can be constructed as the quotient space of $\mathbb{C}^{n+1} \setminus \{0\}$ with the quotient map $(z_0, \dots, z_n) \mapsto [z_0 : \dots : z_n] \in \mathbb{CP}^n$, then f defined above descends to a function $F : \mathbb{CP}^n \rightarrow \mathbb{R}$ defined by $F([z_0 : \dots : z_n]) := f(z_0, \dots, z_n)$.

- (1) Prove that F is a Morse function on $\mathbb{CP}^n$.
- (2) Find all the critical points of F and compute their indices.
- (3) Describe the moduli space $\mathcal{M}(x, y)$ for $x, y \in \text{Crit}(F)$ with the smallest non-zero dimension where the metric is given by the Fubini–Study metric.

Solution. Let $U_k = \{[z_0, \dots, z_n] \in \mathbb{CP}^n \mid z_k \neq 0\}$. Take the standard coordinates of U_k $(w_0, \dots, w_{n-1}) \mapsto [w_0, \dots, w_{k-1}, 1, w_k, \dots, w_{n-1}]$ and let f_k be the function f under the corresponding coordinates. Explicitly,

$$f_k(w_0, \dots, w_{n-1}) = \frac{\sum_{j \leq k-1} j |w_j|^2 + k + \sum_{j \geq k} (j+1) |w_j|^2}{\sum_{j=0}^{n-1} |w_j|^2 + 1}.$$

If $i \leq k-1$, we have

$$\frac{\partial f_k}{\partial x_i} = \frac{2x_j(j-k + \sum_{j \leq k-1} |w_j|^2)}{(\sum_{j=0}^{n-1} |w_j|^2 + 1)^2}, \quad \frac{\partial f_k}{\partial y_i} = \frac{2y_j(j-k - \sum_{j \leq k-1} |w_j|^2)}{(\sum_{j=0}^{n-1} |w_j|^2 + 1)^2}.$$

and if $i \geq k$, we have

$$\frac{\partial f_k}{\partial x_i} = \frac{2x_j(j+1-k + \sum_{j \leq k-1} |w_j|^2)}{(\sum_{j=0}^{n-1} |w_j|^2 + 1)^2}, \quad \frac{\partial f_k}{\partial y_i} = \frac{2y_j(j+1-k - \sum_{j \leq k-1} |w_j|^2)}{(\sum_{j=0}^{n-1} |w_j|^2 + 1)^2}.$$

As a consequence, $\text{Crit}(f) = \{p_0, \dots, p_n\}$, where $p_k = [0, \dots, 1, \dots, 0]$. Moreover, near p_k we have

$$\begin{aligned} f_k(w_0, \dots, w_{n-1}) &= \left(\sum_{j \leq k-1} j |w_j|^2 + k + \sum_{j \geq k} (j+1) |w_j|^2 \right) \left(1 - \sum_{j=0}^{n-1} |w_j|^2 + O(|w|^3) \right) \\ &= k + \left(\sum_{j \leq k-1} (j-k) |w_j|^2 + \sum_{j \geq k} (j+1-k) |w_j|^2 \right) + O(|w|^3), \end{aligned}$$

which implies p_k is non-degenerate.

(2) By (1), $\text{Crit}(f) = \{p_0, \dots, p_n\}$, and we can read out the index of p_k by the local expansion in (1): $\text{ind}(p_k) = 2k$.

(3) $\dim \mathcal{M}(p_k, p_l) = \text{ind } p_k - \text{ind } p_l - 1 = 2k - 2l - 1$ for $k \geq l$. The moduli space with the smallest non-zero dimension is $\mathcal{M}(p_{k+1}, p_k)$, with $\dim \mathcal{M}(p_{k+1}, p_k) = 1$. Since there is no critical point with index $2k+1$, $\partial \mathcal{M}(p_{k+1}, p_k) = \emptyset$ and $\mathcal{M}(p_{k+1}, p_k) = \mathcal{M}(p_{k+1}, p_k)$ is disjoint unions of copies of S^1 . $\square$

Exercise 6.3. Using the fact that the Morse homology is isomorphic to the singular homology, prove the following results for a field $\mathbf{k}$.

- (i) (Poincaré duality) $H_*(M; \mathbf{k}) \cong H_{\dim M - *}(M; \mathbf{k})$;
- (ii) (Künneth formula) $H_*(M \times N; \mathbf{k}) \cong \bigoplus_{i+j=*} H_i(M; \mathbf{k}) \otimes H_j(N; \mathbf{k})$,

where M, N are smooth manifolds.

Solution. (i) Let (F, g) be a Morse–Smale pair. Then $HM_*(F, g; \mathbf{k}) \cong H_*(M; \mathbf{k})$. Consider the pair $(-F, g)$. We have $C_*(F) = C_{\dim M - *}(F)$ and $u(t) \in \widetilde{\mathcal{M}}_{F, g}(x, y)$ if and only if $u(-t) \in \widetilde{\mathcal{M}}_{-F, g}(y, x)$. It follows that $(-F, g)$ is also Morse–Smale and

$$H_*(M; \mathbf{k}) \cong HM_*(F, g; \mathbf{k}) \cong HM_{\dim M - *}(F, g; \mathbf{k}) \cong H_{\dim M - *}(M; \mathbf{k}).$$

(ii) Let $(F, g), (G, h)$ be Morse–Smale pairs on M, N respectively. Then we have $C_i(F) \otimes C_j(G) \cong C_{i+j}(F+G)$ and $(u(t), v(t)) \in \widetilde{\mathcal{M}}_{F+G, g \times h}((x, x'), (y, y'))$ if and only if $u(t) \in \widetilde{\mathcal{M}}_{F, g}(x, y)$ and $v(t) \in \widetilde{\mathcal{M}}_{G, h}(x', y')$. It follows that $(F+G, g \times h)$ is also Morse–Smale and

$$\begin{aligned} H_*(M \times N; \mathbf{k}) &\cong HM_*(F+G, g \times h; \mathbf{k}) \\ &\cong \bigoplus_{i+j=*} HM_i(F, g; \mathbf{k}) \otimes HM_j(G, h; \mathbf{k}) \\ &\cong \bigoplus_{i+j=*} H_i(M; \mathbf{k}) \otimes H_j(N; \mathbf{k}). \end{aligned}$$

$\square$

Exercise 6.4. For the surface $\Sigma_2 = \mathbb{T}^2 \# \mathbb{T}^2$ in the following Figure, take the height function $f : \Sigma_2 \rightarrow \mathbb{R}$ as a Morse function on it. Compute/draw the barcode of the persistence module $\mathbb{V}_f$ coming from the filtered Morse homology groups. Please label the endpoints of bars in the barcode carefully.

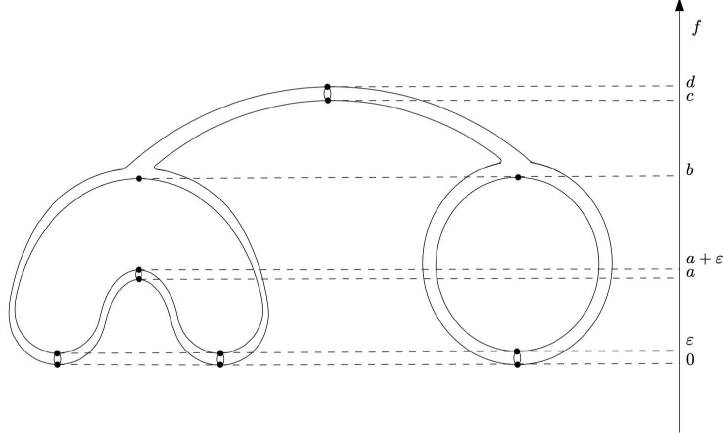


Figure 1: Surface $\Sigma_2 = \mathbb{T}^2 \# \mathbb{T}^2$ with the height function $f : \Sigma_2 \rightarrow \mathbb{R}$

Solution. We first analysis the filtered Morse complex.

$0 < t \leq \varepsilon$, $\{F(x) < t\} = \mathbb{Z}_2^3$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^3$ and $HM_k(\{F(x) < t\}; \mathbb{Z}_2) = 0$ for $k = 1, 2$.

$\varepsilon < t \leq a$, $\{F(x) < t\} = \mathbb{Z}_2^6$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^3$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^3$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = 0$.

$a < t \leq a + \varepsilon$, $\{F(x) < t\} = \mathbb{Z}_2^7$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^2$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^3$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = 0$.

$a + \varepsilon < t \leq b$, $\{F(x) < t\} = \mathbb{Z}_2^8$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^2$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^2$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = 0$.

$b < t \leq c$, $\{F(x) < t\} = \mathbb{Z}_2^{10}$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^2$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^4$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = 0$.

$c < t \leq d$, $\{F(x) < t\} = \mathbb{Z}_2^{11}$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^4$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = 0$.

$t > d$, $\{F(x) < t\} = \mathbb{Z}_2^{12}$. $HM_0(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2$, $HM_1(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2^4$ and $HM_2(\{F(x) < t\}; \mathbb{Z}_2) = \mathbb{Z}_2$.

Denote $\mathbf{k} = \mathbb{Z}_2$. We have

$$\mathcal{B}(\mathbb{V}_f) = \mathbf{k}_{(0,+\infty)} \oplus \mathbf{k}_{(0,a]} \oplus \mathbf{k}_{(0,c]} \oplus \bigoplus_{i=1}^2 \mathbf{k}_{(\varepsilon,+\infty)} \oplus \mathbf{k}_{(\varepsilon,a+\varepsilon]} \oplus \bigoplus_{i=1}^2 \mathbf{k}_{(b,+\infty)} \oplus \mathbf{k}_{(d,+\infty)}$$

□

Exercise 6.5. Prove that, for interval-type persistence $\mathbf{k}$ -modules $\mathbf{k}_{(a,b]}$ and $\mathbf{k}_{(c,d]}$, there exists a non-zero persistence morphism from $\mathbf{k}_{(a,b]}$ to $\mathbf{k}_{(c,d]}$ if and only if $c \leq a$ and $a < d \leq b$.

Exercise 6.6. If $c \leq a$ and $a < d \leq b$,

$$f_t = \begin{cases} 0 & \text{if } t \leq a, t > b \\ 1 & \text{if } a < t \leq b \end{cases}$$

is a persistence module.

Conversely, suppose $f: \mathbf{k}_{(a,b]} \rightarrow \mathbf{k}_{(c,d]}$ is a non-zero persistence morphism, say $f_{t_0} = \mathbb{1}$ for some $a < t_0 \leq b$. Then we must have $c < t_0 \leq d$. In particular, $a < t_0 \leq d$. Then for $a < s < t_0$, $f_{t_0} = i_{s,t_0}^{(c,d]} \circ f_s$, so $i_{s,t_0}^{(c,d]}$ is non-zero for $a < s < t_0$ and then $c < s$. Taking $s \rightarrow a^+$ yields $c \leq a$. Similarly, for $t_0 < s \leq d$, $f_{t_0} = f_s \circ i_{t_0,s}^{(a,b]}$, so $i_{t_0,s}^{(a,b]}$ is non-zero for $t_0 < s \leq d$ and then $s \leq b$. Taking $s \rightarrow d^-$ yields $d \leq b$.

Exercise 6.7. Prove that the Arnold conjecture holds for any time-1 map φ_H^1 of a Hamiltonian flow that is C^1 -close to the identity map.

Solution. Let φ_H^t be a Hamiltonian flow that is C^1 -close to the identity map, with corresponding Hamiltonian vector field denoted by X_H^t . Then the graph of φ_H^t is a Lagrangian submanifold in $(M \times M, \omega \oplus -\omega)$ which is close to the diagonal Δ . Also a neighbourhood of Δ may be identified with a neighborhood of the zero section in $(T^*M, d\lambda_{\text{can}})$. As the flow is C^1 -close to the identity map, we can find a uniform neighborhood of the zero section containing the graph for every φ_H^t . Then

$$\frac{d}{dt}(\varphi_H^t)^* \lambda_{\text{can}} = (\varphi_H^t)^* \mathcal{L}_{X_H^t} \lambda = (\varphi_H^t)^* (d\iota_{X_H^t} \lambda_{\text{can}} - dH) = d(\varphi_H^t)^* (\iota_{X_H^t} \lambda_{\text{can}} - H),$$

which implies

$$(\varphi_H^t)^* \lambda_{\text{can}} = \lambda_{\text{can}} + \int_0^t d(\varphi_H^s)^* (\iota_{X_H^s} \lambda_{\text{can}} - H) ds = \lambda_{\text{can}} + d \int_0^t (\varphi_H^s)^* (\iota_{X_H^s} \lambda_{\text{can}} - H) ds.$$

We see that $(\varphi_H^t)^* \lambda_{\text{can}} = \lambda_{\text{can}} + dF_t$, where

$$F_t = \int_0^t (\varphi_H^s)^* (\iota_{X_H^s} \lambda_{\text{can}} - H) ds.$$

We set $F = F_1$. It follows that the graph of $(\varphi_H^1)^* \lambda_{\text{can}} - \lambda_{\text{can}}$ is $\varphi_H^1(M)$, so the fixed points of φ^1 are the critical points of F , and that an element $p \in \text{Fix}(\varphi) = \text{Crit}(F)$ is nondegenerate as a fixed point of φ if and only if it is nondegenerate as a critical point of F . $\square$

Exercise 6.8. Given a contact manifold $(M, \xi = \ker \alpha)$ with a fixed contact 1-form α . Consider the following action functional $\mathcal{A}_\alpha: \Lambda M \rightarrow \mathbb{R}$ defined by

$$\gamma \mapsto \mathcal{A}_\alpha(\gamma) := \int_\gamma \alpha.$$

Prove that $\gamma \in \text{Crit}(\mathcal{A}_\alpha)$ if and only if γ is a closed Reeb orbit with period $\mathcal{A}_\alpha(\gamma)$.

Solution. To compute the variation of $\mathcal{A}_\alpha(\gamma)$ with respect to $\xi \in \Gamma(\gamma^* TM)$, we extend $\gamma(t)$ to a variation $f(s, t) = f_s(t)$ such that

$$f_0(t) = \gamma(t), \quad \frac{\partial f_s}{\partial s}(t) = \xi(t).$$

Taking variation at $s = 0$, we obtain

$$\begin{aligned}
\left. \frac{d}{ds} \right|_{s=0} \mathcal{A}_\alpha(f_s(t)) &= \int_0^1 \left. \frac{d}{ds} \right|_{s=0} f_s^* \alpha dt \\
&= \int_0^1 \gamma^*(\mathcal{L}_\xi \alpha) \\
&= \int_0^1 \gamma^*(d\iota_\xi \alpha + \iota_\xi d\alpha) \\
&= \int_0^1 d(\gamma^* \iota_\xi \alpha) + \int_\gamma \iota_\xi d\alpha.
\end{aligned}$$

By Stokes' formula, the first term vanishes, hence

$$\left. \frac{d}{ds} \right|_{s=0} \mathcal{A}_\alpha(f_s(t)) = \int_0^1 d\alpha(\xi(\gamma(t)), \dot{\gamma}(t)) dt.$$

We see that $\gamma \in \text{Crit}(\mathcal{A}_\alpha)$ if and only if $d\alpha(\xi(\gamma(t)), \dot{\gamma}(t)) = 0$ for all $\xi \in \Gamma(\gamma^* TM)$, if and only if $\dot{\gamma}(t)$ is a multiple of the Reeb vector field, if and only if γ is a closed Reeb orbit with period

$$\mathcal{A}_\alpha(\gamma) = \int_\gamma \alpha = \int_0^1 \alpha(\dot{\gamma}(t)) dt.$$

□

Exercise 6.9. Consider a Morse function $F : M \rightarrow \mathbb{R}$ for a symplectic manifold (M, ω) . Viewing F as a time-independent Hamiltonian function, a critical point $p \in \text{Crit}(F)$ corresponds to a closed Hamiltonian orbit which is the constant orbit at p . Prove that if p is non-degenerate as a closed Hamiltonian orbit, then it is a non-degenerate critical point of F . Is the converse statement above true?

Solution. By Darboux theorem, we can pick local coordinates near p such that $\omega = \omega_{\text{std}}$. Let φ^t be the Hamiltonian flow generated by F . Direct computation yields

$$\frac{\partial}{\partial t} (\text{Jac}_p(\varphi^t)) = \text{Jac}_p \left(\frac{\partial}{\partial t} \varphi^t \right) = \text{Jac}_p (X_F \circ \varphi^t).$$

Here we regard a vector field as a local smooth function $U \rightarrow \mathbb{R}^{2n}$. By definition, $X_F = J_0 \text{Jac}(F)$, which implies

$$\frac{\partial}{\partial t} (\text{Jac}_p \varphi^t) = \text{Jac}_p (J_0 \text{Jac}(F) \circ \varphi^t) = (J_0 \text{Hess}_{\varphi^t(p)}(F)) \cdot \text{Jac}_p \varphi^t.$$

The matrix of $d\varphi_p^t$ writes

$$\exp(t J_0 \text{Hess}_{\varphi^t(p)}(F)).$$

Now we assume that p is a critical point, so $\varphi^t(p) = p$. If p is a non-degenerate closed Hamiltonian orbit, then $\text{Hess}_p(F)$ admits no eigenvalues in $2\pi\mathbb{Z}$ and in particular, F is non-degenerate at p .

However, we can also see that the converse is not true: $\text{Hess}_p(F)$ may have nonzero eigenvalues in $2\pi\mathbb{Z}$, making 1 an eigenvalue of $d(\varphi_H^1)_p$. For example, on $\mathbb{R}^2$, the Hamiltonian flow generated by a Morse function $F = \pi(x^2 + y^2)$ is $\varphi^t(x, y) = (x \cos(2\pi t) - y \sin(2\pi t), x \sin(2\pi t) + y \cos(2\pi t))$. 0 is a non-degenerate critical point of F , but $\varphi^1 = \mathbb{1}$ and 0 is degenerate as a closed Hamiltonian orbit. $\square$

Exercise 6.10. On $(\mathbb{R}^2, \omega_{\text{std}})$, consider a Hamiltonian function $H(x, y) = \pi\alpha(x^2 + y^2)$ with $\alpha \in \mathbb{R} \setminus \mathbb{Z}$. Viewing 0 as a closed Hamiltonian orbit of the Hamiltonian flow $\{\varphi_H^t\}_{t \in [0, 1]}$, compute its Conley-Zehnder index $\mu_{\text{CZ}}(0)$ and write it as an expression in terms of α .

Solution. The Hamiltonian flow generated by H is

$$\varphi^t(x, y) = (x \cos(2\pi\alpha t) - y \sin(2\pi\alpha t), x \sin(2\pi\alpha t) + y \cos(2\pi\alpha t)).$$

Under the basis ∂_x, ∂_y , $(d\varphi^t)_0$ writes

$$A(t) = \begin{pmatrix} \cos(2\pi\alpha t) & \sin(2\pi\alpha t) \\ -\sin(2\pi\alpha t) & \cos(2\pi\alpha t) \end{pmatrix}.$$

We have

$$\dot{A}(t) = 2\pi\alpha \begin{pmatrix} -\sin(2\pi\alpha t) & \cos(2\pi\alpha t) \\ -\cos(2\pi\alpha t) & -\sin(2\pi\alpha t) \end{pmatrix}.$$

Since $A(t)$ is orthogonal,

$$J(t) = -J_0 \dot{A}(t) A(t)^{-1} = -J_0 \dot{A}(t) A(t)^T = -2\pi\alpha I_2.$$

Since $\varphi^t = \mathbb{1}$ for $t \in \alpha^{-1}\mathbb{Z}$, $\ker(\mathbb{1} - \varphi^t) = \mathbb{R}^2$. As a consequence,

$$\mu_{\text{CZ}}(0) = -\text{sgn}(\alpha)(1 + 2\#\{t \in [0, 1] \mid \alpha t \in \mathbb{Z}\}).$$

$\square$

Exercise 6.11. Consider a path of matrices $A = \{A(t)\}_{t \in [0, 1]}$ given by the following symplectic shear matrices,

$$A(t) = \begin{pmatrix} I_n & B(t) \\ 0 & I_n \end{pmatrix} \in \text{Sp}(2n)$$

where $B(t)$ is an $n \times n$ symmetric matrix. Prove that $\mu_{\text{RS}}(A) = \frac{1}{2}(\text{sign}(B(0)) - \text{sign}(B(1)))$. (Hint: The Robbin-Salamon index is invariant under taking conjugation by an element of $\text{Sp}(2n)$ and homotopies with fixed end points.)

Solution. As the index is invariant under path homotopies, we may assume that $B(t)$ has distinct eigenvalues for all t . Therefore, we can find a smooth path of orthogonal matrices $Q(t)$ such that $Q(t)^T B(t) Q(t) = D(t)$ with $D(t)$ diagonal. Then

$$\begin{pmatrix} Q(t) & \\ & Q(t)^T \end{pmatrix}^{-1} A(t) \begin{pmatrix} Q(t)^T & \\ & Q(t) \end{pmatrix} = \begin{pmatrix} I_n & D(t) \\ 0 & I_n \end{pmatrix}.$$

As a consequence, we can reduce the case to $n = 1$.

Consider the path

$$\Psi(t) = \begin{pmatrix} a(t) & D(t) \\ c(t) & E(t) = a(t)^{-1}(1 + c(t)D(t)) \end{pmatrix},$$

where $a(t), c(t)$ are smooth functions on $[0, 1]$ with $a(0) = a(1) = 0, c(0) = c(1) = 0$. Direct computation yields

$$\begin{aligned} S(t) &= -J_0 \dot{\Psi}(t) \Psi(t)^{-1} \\ &= \begin{pmatrix} \dot{c}E - c\dot{E} & \dot{D}c - \dot{a}E \\ \dot{D}c - \dot{a}E & \dot{a}D - a\dot{D} \end{pmatrix}, \end{aligned}$$

hence

$$\begin{aligned} S(0) &= \begin{pmatrix} \dot{c}(0) & -\dot{a}(0) \\ -\dot{a}(0) & \dot{a}(0)D(0) - \dot{D}(0) \end{pmatrix}, \\ S(1) &= \begin{pmatrix} \dot{c}(1) & -\dot{a}(1) \\ -\dot{a}(1) & \dot{a}(1)D(1) - \dot{D}(1) \end{pmatrix}. \end{aligned}$$

Take a such that $\dot{a}(0), \dot{a}(1)$ are non-zero, and $a(t) > 1$ for $0 < t < 1$. Take c such that $c(t)D(t) \geq 0$. Then the only crossings of $\Psi(t)$ are 0, 1. If $D(0) = 0$, $\ker(\mathbb{1} - \Psi(0)) = \mathbb{R}^2$. If $D(0) \neq 0$, $\ker(\mathbb{1} - \Psi(0))$ is spanned by $(1, 0)^T$. The same holds for $D(1)$.

Take c such that

$$\begin{cases} \dot{c}(0) = 0 \text{ if } D(0) = 0, \\ \dot{c}(0) = \text{sign}(D(0)) \text{ if } D(0) \neq 0, \\ \dot{c}(1) = 0 \text{ if } D(1) = 0, \\ \dot{c}(1) = -\text{sign}(D(1)) \text{ if } D(1) \neq 0. \end{cases}$$

For $t_0 = 0, 1$, if $D(t_0) \neq 0$, then $\Gamma(\Psi, t_0) = (\dot{c}(t_0))$ and if $D(t_0) = 0$, then

$$\Gamma(\Psi, t_0) = \begin{pmatrix} 0 & -\dot{a}(t_0) \\ -\dot{a}(t_0) & -\dot{D}(t_0) \end{pmatrix}.$$

By our choice of a, c , both 0, 1 are regular crossings and

$$\begin{aligned} \text{sign}(\Gamma(\Psi, 0)) &= \text{sign}(D(0)) = \text{sign}(B(0)) \\ \text{sign}(\Gamma(\Psi, 1)) &= -\text{sign}(D(1)) = -\text{sign}(B(1)). \end{aligned}$$

The conclusion follows. □

7 HOMEWORK6

Exercise 7.1. Prove that d_{Hofer} satisfies the following properties for any $\varphi, \psi, \theta \in \text{Ham}(M, \omega)$:

- (1) $d_{\text{Hofer}}(\varphi, \psi) \geq 0$.
- (2) $d_{\text{Hofer}}(\varphi, \psi) = d_{\text{Hofer}}(\psi, \varphi)$.
- (3) $d_{\text{Hofer}}(\varphi, \theta) \leq d_{\text{Hofer}}(\varphi, \psi) + d_{\text{Hofer}}(\psi, \theta)$.
- (4) $d_{\text{Hofer}}(\theta\varphi, \theta\psi) = d_{\text{Hofer}}(\varphi\theta, \psi\theta) = d_{\text{Hofer}}(\varphi, \psi)$.

Solution. (1) is obvious by definition. Recall that if $\{f_t\}, \{g_t\}$ are the Hamiltonian flows generated by normalized Hamiltonians F_t, G_t , respectively, then the flow $h_t = f_t g_t$ is generated by the normalized Hamiltonian $H_t = F_t + G_t \circ f_t^{-1}$. Note that this conclusion does not depend on f_0, g_0 , so they holds for Hamiltonian paths. f_t is a path from φ to ψ if and only if $\varphi f_t^{-1} \psi$ is a path from ψ to φ , so a normalized Hamiltonian F_t generates f_t if and only if $-F_t \circ (f_t \varphi^{-1})$ generates $\varphi f_t^{-1} \psi$. Note that $-F_t \circ (f_t \varphi^{-1})$ is normalized if and only if F_t is normalized.

As

$$\begin{aligned} \|-F_t \circ (f_t \varphi^{-1})\|_{\text{Hofer}} &= \int_0^1 (\max_M (-F_t \circ (f_t \varphi^{-1})) - \min_M (-F_t \circ (f_t \varphi^{-1}))) dt \\ &= \int_0^1 (\max_M F_t - \min_M F_t) dt \\ &= \|F_t\|_{\text{Hofer}}, \end{aligned}$$

we obtain (2).

Similarly, f_t is a path from φ to ψ if and only if θf_t is a path from $\theta\varphi$ to $\theta\psi$, so a normalized Hamiltonian F_t generates f_t if and only if $F_t \circ \theta^{-1}$ generates θf_t . f_t is a path from φ to ψ if and only if $f_t \theta$ is a path from $\varphi\theta$ to $\psi\theta$, and $f_t, f_t \theta$ can be generated by the same Hamiltonian. Since $\|F_t \circ \theta\|_{\text{Hofer}} = \|F_t\|_{\text{Hofer}}$, we obtain (4).

Let $\{f_t\}, \{g_t\}$ be the Hamiltonian flows generated by normalized Hamiltonians F_t, G_t , respectively. Then the normalized Hamiltonian $H_t = F_t + G_t \circ f_t^{-1}$ generates $h_t = f_t g_t$.

$$\begin{aligned} \|H_t\|_{\text{Hofer}} &= \int_0^1 (\max_M H_t - \min_M H_t) dt \\ &\leq \int_0^1 ((\max_M F_t + \max_M G_t \circ f_t^{-1}) - (\min_M F_t + \min_M G_t \circ f_t^{-1})) dt \\ &= \int_0^1 ((\max_M F_t + \max_M G_t) - (\min_M F_t + \min_M G_t)) dt \\ &= \|F_t\|_{\text{Hofer}} + \|G_t\|_{\text{Hofer}}. \end{aligned}$$

Fix $f_1 = \varphi^{-1} \psi, g_1 = \psi^{-1} \theta$, we obtain

$$d_{\text{Hofer}}(\mathbf{1}, \varphi^{-1} \theta) \leq \|H_t\|_{\text{Hofer}} \leq \|F_t\|_{\text{Hofer}} + \|G_t\|_{\text{Hofer}}.$$

Taking infimum with respect to F_t, G_t , we get

$$d_{\text{Hofer}}(\mathbb{1}, \varphi^{-1}\theta) \leq d_{\text{Hofer}}(\mathbb{1}, \varphi^{-1}\psi) + d_{\text{Hofer}}(\mathbb{1}, \psi^{-1}\theta).$$

Now (3) follows from (4). $\square$

Exercise 7.2. For any $\varphi, \psi \in \text{Ham}(M, \omega)$, we define

$$d_{L^p}(\varphi, \psi) := \inf \left\{ \int_0^1 \left(\int_M |H_t|^p d\text{vol} \right)^{\frac{1}{p}} dt \mid \begin{array}{l} H \text{ is a normalized Hamiltonian} \\ \text{generating a path from } \varphi \text{ to } \psi \end{array} \right\},$$

where the generation means that there exists $\{\theta_t\}_{t \in [0,1]} \subset \text{Ham}(M, \omega)$ with $\theta_0 = \varphi$ and $\theta_1 = \psi$ such that $\frac{d\theta_t}{dt} = X_{H_t}(\theta_t)$. Prove that $d_{L^p} \equiv 0$ for $p \in \mathbb{Z}_{\geq 1}$.

Solution. It suffices to show that the L^p -displacement energy of a open set vanishes. We work with the Darboux coordinates, so assume the symplectic form is given by ω_{std} . We can find $r > 0$ such that $U = B(0, r)$ is contained in the chart. Let $A \subset U$ be the ball with the same center of the radius $\frac{1}{4}r$. Consider a local Hamiltonian flow h_t generated by $H(x, y) = r \cdot x_1$. Clearly, $h_1(A) \cap A = \emptyset$. Denote by S_t the sphere $h_t(\partial A)$. Let F_t be a time-dependent smooth function, obtained from H by a cut off outside a small neighborhood of S_t , and take $G_t = F_t - c_t$, where $c_t = 0$ if M is open and $c_t = \text{vol}(M)^{-1} \int_M F_t d\text{vol}$ if M is closed. Then the Hamiltonian flow generated by the normalized Hamiltonian G_t satisfies $g_t(A) = h_t(A)$. For $p \geq 1$ and every $\varepsilon > 0$, we can take appropriate cutoffs such that

$$\int_0^1 \left(\int_M |G_t|^p d\text{vol} \right)^{\frac{1}{p}} dt = \int_0^1 \left(\int_U |G_t|^p d\text{vol} \right)^{\frac{1}{p}} dt \leq \int_0^1 \left(\int_{S_t} |G_t|^p d\text{vol} \right)^{\frac{1}{p}} dt + \varepsilon = \varepsilon,$$

hence the L^p -displacement energy of A vanishes. $\square$

Exercise 7.3. Suppose a group G is equipped with a bi-invariant pseudo-metric denoted by d . Prove that for any elements $w, x, y, z \in G$, we have

$$d(e_G, [w, x]) - d(e_G, [y, z]) \leq 2(d(w, y) + d(x, z)),$$

where $[x, y] = xyx^{-1}y^{-1}$.

Solution. We have

$$\begin{aligned} d(e_G, [w, x]) - d(e_G, [y, z]) &\leq d(e_G, [w, x][y, z]^{-1}) \\ &= d(e_G, wxw^{-1}x^{-1}zyz^{-1}y^{-1}) \\ &= d(wx^{-1}w^{-1}zyz^{-1}, x^{-1}z) \\ &\leq d(wx^{-1}w^{-1}zyz^{-1}, e_G) + d(e_G, x^{-1}z) \\ &= d(x, z) + d(w^{-1}y, xw^{-1}yz^{-1}) \\ &\leq d(x, z) + d(w^{-1}y, e_G) + d(e_G, xw^{-1}yz^{-1}) \\ &= d(x, z) + d(w, y) + d(x^{-1}z, w^{-1}y) \\ &\leq 2(d(w, y) + d(x, z)). \end{aligned}$$

$\square$

Exercise 7.4. Let G be a group equipped with a conjugate invariant norm $\|\cdot\|$. Consider a set S equipped with a transitive group action of G . The conjugate invariant norm $\|\cdot\|$ induces a pseudo-metric $\delta_{\|\cdot\|}$ on S , defined by

$$\delta_{\|\cdot\|}(s_1, s_2) := \inf\{\|g\| \mid gs_1 = s_2\},$$

for any $s_1, s_2 \in S$. Fix a basepoint $s_0 \in S$ and define $H := \{g \in G \mid gs_0 = s_0\}$. Prove that the closure of H with respect to the topology on G induced by $\|\cdot\|$ is a subgroup. In particular, $\delta_{\|\cdot\|}$ is non-degenerate if and only if H is closed.

Solution. H is the stabilizer of $s_0 \in S$, so H is a subgroup of G . G is a topological group with respect to the metric topology induced by $\|\cdot\|$, and it is a general result that the closure of a subgroup of a topological group is still a subgroup.

Suppose H is closed. Assume that $\delta_{\|\cdot\|}(s_1, s_2) = 0$ and we need to show that $s_1 = s_2$. By definition, we can pick a sequence $\{g_i\}$ from G with $g_i s_1 = s_2$ and $\|g_i\| \rightarrow 0$. Since the group action is transitive, $s_1 = h_1 s_0$, $s_2 = h_2 s_0$ for some $h_1, h_2 \in G$. Then $h_2^{-1} g_i h_1 \in H$ with $d(h_2^{-1} g_i h_1, h_2^{-1} h_1) = \|h_2^{-1} g_i h_1\| = \|g_i\| \rightarrow 0$. As H is closed, $h_2^{-1} h_1 \in H$ and then $s_1 = s_2$.

Conversely, suppose $\delta_{\|\cdot\|}(s_1, s_2)$ is non-degenerate. Assume a sequence $\{g_i\}$ in H converges to $g \in G$, and we need to show that $g \in H$. Let $s_1 = g s_0$. Then $g g_i^{-1} s_0 = g s_0 = s_1$, so $\delta_{\|\cdot\|}(s_0, s_1) \leq \|g g_i^{-1}\| = d(g, g_i) \rightarrow 0$. It follows that $s_1 = s_0$ and then $g \in H$. $\square$

Exercise 7.5. Let $\mathbb{B}_1, \mathbb{B}_2$ be two barcodes. Let $a_1 \geq \dots \geq a_n$ be the first n smallest left endpoints of intervals in $\mathbb{B}_1$, and the corresponding intervals are $(a_1, c_1], \dots, (a_n, c_n]$ where c_i could be $+\infty$. Similarly, $b_1 \geq \dots \geq b_n$ be the first n smallest left endpoints of intervals in $\mathbb{B}_2$, and the corresponding intervals are $(b_1, d_1], \dots, (b_n, d_n]$. If $\min\{c_1, \dots, c_n, d_1, \dots, d_n\} \geq \max\{a_1, b_1\}$, then

$$\frac{1}{2} \max_{1 \leq i \leq n} |a_i - b_i| \leq d_{\text{bottleneck}}(\mathbb{B}_1, \mathbb{B}_2).$$

Solution. By contradiction, suppose $\max |a_i - b_i| > 2d_{\text{bottleneck}}(\mathbb{B}_1, \mathbb{B}_2)$. Then there is a δ -matching such that $2\delta < |a_j - b_j|$. This matching gives rise to a permutation $\sigma \in S_n$ if we add an extra correspondence between the erased bars. If $(a_i, c_i]$ is erased, then $a_i - b_{\sigma(i)} \leq d_{\sigma(i)} - b_{\sigma(i)} \leq 2\delta$ and similarly $b_{\sigma(i)} - a_i \leq c_i - a_i \leq 2\delta$. We see that for all i , $|a_i - b_{\sigma(i)}| \leq 2\delta$ holds. By the matching lemma,

$$\max_i |a_i - b_i| = \min_{\tau \in S_n} \max_i |a_i - b_{\tau(i)}| \leq \max_i |a_i - b_{\sigma(i)}| \leq 2\delta,$$

a contradiction. $\square$

Exercise 7.6. Let $\mathbb{V}$ be a persistence module equipped with a persistence morphism $J: \mathbb{V} \rightarrow \mathbb{V}$, where $J = \{J_t: V_t \rightarrow V_t\}_{t \in \mathbb{R}}$ such that $J_t^2 = -\mathbf{1}$. For any interval $I \subset \mathbb{R}$, we define

$$m(I, \mathbb{B}(\mathbb{V})) = \#\{\text{bars in } \mathbb{B}(\mathbb{V}) \text{ that contains } I\}.$$

Prove that for any $I \subset \mathbb{R}$, the number $m(I, \mathbb{B}(\mathbb{V}))$ is always even.

Solution. Let $I = (a, b]$ and take $s \in I$. We can take s sufficiently close to a such that $m(I, \mathbb{B}(\mathbb{V}))$ coincides with $\dim \text{im } i_{s,b}$. Since $i_{s,b} J_s = J_b i_{s,b}$, J_b restricts to a complex structure on $\text{im } i_{s,b}$, which implies $m(I, \mathbb{B}(\mathbb{V})) = \dim \text{im } i_{s,b}$ is even. $\square$

8 HOMEWORK 7

Exercise 8.1. Work out the definition of $\iota_{r,s}: \text{SH}_*^{(e^r, \infty)}(W) \rightarrow \text{SH}_*^{(e^s, \infty)}(W)$ for $r \leq s$.

Solution. We first fix $H \in \mathcal{H}(W)$. Since the action decreases along the Floer trajectories, the chain map $\text{CF}_*^{(-\infty, e^r)}(H) \rightarrow \text{CF}_*^{(-\infty, e^s)}(H)$ is well-defined, so we get a well-defined chain map $\text{CF}_*^{(e^r, \infty)}(H) \rightarrow \text{CF}_*^{(e^s, \infty)}(H)$. The chain map induces $\text{HF}_*^{(e^r, \infty)}(H) \rightarrow \text{HF}_*^{(e^s, \infty)}(H)$, and we obtain $\text{SH}_*^{(e^r, \infty)}(W) \rightarrow \text{HF}_*^{(e^s, \infty)}(H)$ via the canonical map $\text{SH}_*^{(e^r, \infty)}(W) \rightarrow \text{HF}_*^{(e^r, \infty)}(H)$. Moreover, the following digram commutes

$$\begin{array}{ccc} & \text{SH}_*^{(e^r, \infty)}(W) & \\ \swarrow & & \searrow \\ \text{HF}_*^{(e^s, \infty)}(H) & \xrightarrow{\quad\quad\quad} & \text{HF}_*^{(e^s, \infty)}(G) \end{array}$$

for $G \leq H$ because the following digram commutes

$$\begin{array}{ccc} \text{HF}_*^{(e^r, \infty)}(H) & \xrightarrow{\quad\quad\quad} & \text{HF}_*^{(e^r, \infty)}(G) \\ \downarrow & & \downarrow \\ \text{HF}_*^{(e^s, \infty)}(H) & \xrightarrow{\quad\quad\quad} & \text{HF}_*^{(e^s, \infty)}(G) \end{array}$$

for $H \leq G$. By the universal property of inverse limit, we obtain a morphism $\text{SH}_*^{(e^r, \infty)}(W) \rightarrow \text{SH}_*^{(e^s, \infty)}(W)$. $\square$

Exercise 8.2. Prove that any symplectic capacity c satisfies $c \geq w_G$ in the sense that $c(U) \geq w_G(U)$ for any Liouville domain U in $\mathbb{R}^{2n}$. In other words, within all symplectic capacities, the Gromov width w_G is the smallest one. Is there a largest symplectic capacity?

Solution. Let c be such a symplectic capacity. Then for every $B^{2n}(r)$ admitting a symplectic embedding into U , $r = c(B^{2n}(r)) \leq c(U)$. Taking the superium with respect to r we obtain $w_G(U) \leq c(U)$.

Define $c_Z = \inf \{R \geq 0 \mid \text{there exists a symplectic embedding } U \rightarrow Z^{2n}(R)\}$ (if there is no such symplectic embedding, c_Z is ∞). We show that c_Z is indeed a symplectic capacity. Indeed, a symplectic embedding $U \rightarrow V$ implies $c_Z(U) \leq c_Z(V)$ and $c_Z(\lambda U) = \lambda c_Z(U)$ for $\lambda > 0$. Gromov's non-squeezing theorem implies that $c_Z(B^{2n}(r)) = r$. Suppose c is a symplectic capacity. Then for every $Z^{2n}(R)$ admitting a symplectic embedding $U \rightarrow Z^{2n}(R)$, $c(U) \leq c(Z^{2n}(R)) = R$. Taking the infimum with respect to R we obtain $c(U) \leq c_Z(U)$. $\square$

Exercise 8.3. Compute the limit

$$\lim_{k \rightarrow +\infty} \frac{c_k^{\text{ECH}}(E(a, b))^2}{k}$$

and explain this result in a geometric way.

Solution. Denote $c_k^{\text{ECH}}(E(a, b))$ by c_k . $\{(x, y) \in \mathbb{R}_{\geq 0}^2 \mid ax + by \leq c_k\}$ contains $k + 1$ lattice points. By assigning each lattice points lying in the triangle to the square for which it serves as the lower-left corner, we cover the triangle, thus

$$\frac{c_k^2}{2ab} \leq k + 1.$$

On the other hand, by assigning each lattice points lying in the triangle to the square for which it serves as the upper-right corner, we obtain

$$\frac{c_k^2}{2ab} \geq k - \lfloor \frac{c_k}{a} \rfloor - \lfloor \frac{c_k}{b} \rfloor.$$

By the first equality we have $c_k = o(k)$, so

$$\lim_{k \rightarrow +\infty} \frac{c_k^{\text{ECH}}(E(a, b))^2}{k} = 2ab.$$

□

9 HOMEWORK 8

Exercise 9.1. With the notations in the lecture note of Section 7.1, for any $\lambda \in \mathbb{R}$, prove that the following diagrams commute,

$$\begin{array}{ccccc} \mathbb{V}_{H_1, \lambda} & \xrightarrow{\sigma_{H_1, H_2}} & \mathbb{V}_{H_2, \lambda + \int_0^1 \max_M (H_{2,t} - H_{1,t}) dt} & \xrightarrow{\sigma_{H_2, H_1}} & \mathbb{V}_{H_1, \lambda + \|H_2 - H_1\|_{\text{Hofer}}} \\ & \searrow & & \nearrow & \\ & & \mathbb{V}_{H_1} & & \\ & & \iota_{\lambda, \lambda + \|H_2 - H_1\|_{\text{Hofer}}} & & \end{array}$$

and

$$\begin{array}{ccccc} \mathbb{V}_{H_2, \lambda} & \xrightarrow{\sigma_{H_2, H_1}} & \mathbb{V}_{H_1, \lambda + \int_0^1 \max_M (H_{1,t} - H_{2,t}) dt} & \xrightarrow{\sigma_{H_2, H_1}} & \mathbb{V}_{H_2, \lambda + \|H_2 - H_1\|_{\text{Hofer}}} \\ & \searrow & & \nearrow & \\ & & \mathbb{V}_{H_2} & & \\ & & \iota_{\lambda, \lambda + \|H_2 - H_1\|_{\text{Hofer}}} & & \end{array}$$

where $\|H_2 - H_1\|_{\text{Hofer}}$ denotes the Hofer's norm of Hamiltonian function $H_2 - H_1$, and $\iota_{\cdot, \cdot}$ denotes a map induced by the inclusion of the filtered chain complexes.

Solution. Since

$$\begin{aligned} \|H_2 - H_1\|_{\text{Hofer}} &= \int_0^1 \max_M (H_{2,t} - H_{1,t}) dt + \int_0^1 \min_M (H_{2,t} - H_{1,t}) dt \\ &= \int_0^1 \max_M (H_{2,t} + H_{1,t}) dt - \int_0^1 \max_M (H_{1,t} - H_{2,t}) dt, \end{aligned}$$

the composition $\sigma_{H_2, H_1} \circ \sigma_{H_1, H_2}$ is well-defined. Since $\sigma_{H_2, H_1} \circ \sigma_{H_1, H_2}: \text{HF}_*(H_1) \rightarrow \text{HF}_*(H_2) \rightarrow \text{HF}_*(H_1)$ is the identity, we see that $\sigma_{H_2, H_1} \circ \sigma_{H_1, H_2} = \mathbb{V}_{H_1} \iota_{\lambda, \lambda + \|H_2 - H_1\|_{\text{Hofer}}}$.

The same argument works for $\sigma_{H_1, H_2} \circ \sigma_{H_2, H_1}$. $\square$

Exercise 9.2. (i) Let X_Ω be a toric concave domain with the profile curve defined by $\{(x_1, x_2) \in \mathbb{R}_{\geq 0}^2 \mid (x_1)^{\frac{2}{3}} + (x_2)^{\frac{2}{3}} = 1\}$. Prove that X_Ω is a (geometrically) convex domain in $\mathbb{C}^2$ (though the moment image Ω is *not* a convex subset in $\mathbb{R}_{\geq 0}^2$)

(ii) In general, prove that X_Ω is a (geometrically) convex domain in $\mathbb{C}^2$ if and only if the set

$$\sqrt{\Omega} = \{(p_1, p_2) \in \mathbb{R}^2 \mid (|p_1|^2, |p_2|^2) \in \Omega\}$$

is a convex subset in $\mathbb{R}^2$.

Solution. (i) We have

$$X_\Omega = \{(z_1, z_2) \in \mathbb{C}^2 \mid |z_1|^3 + |z_2|^3 \leq 1\}.$$

Note that $z \mapsto |z|^3$ is a convex function, so $|tz_1 + (1-t)z'_1|^3 \leq t|z_1|^3 + (1-t)|z'_1|^3$ for $t \in [0, 1]$. It follows that

$$|tz_1 + (1-t)z'_1|^3 + |tz_2 + (1-t)z'_2|^3 \leq t(|z_1|^3 + |z_2|^3) + (1-t)(|z'_1|^3 + |z'_2|^3) = 1$$

for $(z_1, z_2), (z'_1, z'_2) \in X_\Omega$ and $t \in [0, 1]$.

(ii) $(z_1, z_2) \in X_\Omega$ if and only if $(|z_1|^2, |z_2|^2) \in \Omega$, if and only if $(|z_1|, |z_2|) \in \sqrt{\Omega}$.

Assume that $\sqrt{\Omega}$ is convex. Then for $(z_1, z_2), (z'_1, z'_2) \in X_\Omega$ and $t \in [0, 1]$,

$$t(|z_1|, |z_2|) + (1-t)(|z'_1|, |z'_2|) \in \sqrt{\Omega}.$$

Since $|tz_1 + (1-t)z'_1| \leq t|z_1| + (1-t)|z'_1|$, $|tz_2 + (1-t)z'_2| \leq t|z_2| + (1-t)|z'_2|$, we have $(|tz_1 + (1-t)z'_1|^2, |tz_2 + (1-t)z'_2|^2) \in \Omega$, which implies $t(z_1, z_2) + (1-t)(z'_1, z'_2) \in X_\Omega$.

Conversely, assume that X_Ω is convex. Note that $\sqrt{\Omega} = X_\Omega \cap \mathbb{R}_{\geq 0}^2$, so $\sqrt{\Omega}$ is convex. $\square$

Exercise 9.3. Prove that for two star-shaped domains $X, Y \subset (\mathbb{R}^4, d\lambda_{\text{std}})$, if there exists a symplectic embedding $\varphi: X \rightarrow Y$, then $\text{Sh}^+(X) \subset \text{Sh}^+(Y)$.

Solution. Let (e_1, e_2) be the ordered basis of $H_1(L; \mathbb{Z})$ satisfying the proposition in lecture notes. It suffices to show that $(\varphi_*e_1, \varphi_*e_2)$ is the ordered basis of $H_1(\varphi(L); \mathbb{Z})$ satisfying the proposition.

To start with, $\varphi(L) \subset Y$ is still a Lagrangian torus and $\varphi_*e_1, \varphi_*e_2$ is indeed a basis. For $\gamma: S^1 \rightarrow L$, we have $\mu(\gamma) = \mu(\varphi \circ \gamma)$. For any 2-chain Σ in X such that $\partial\Sigma = e_1$, we have $\partial\varphi_*(\Sigma) = \varphi_*\partial\Sigma = \varphi_*e_1$, hence

$$\Omega(\varphi_*e_1) = \int_{\varphi_*\Sigma} (d\lambda_{\text{std}}) = \int_{\Sigma} \varphi^*(d\lambda_{\text{std}}) = \int_{\Sigma} d\lambda_{\text{std}} = \Omega(e_1).$$

Similarly, $\Omega(e_2) = \Omega(\varphi_*e_2)$. $\square$